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Comparative study of effectiveness of *Eichhornia*, *Acorus* & *Ipomoea* against Coleopteran pest in mixed ratio

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Abstract- Leaf powder of *Eichhornia*, *Acorus* & *Ipomoea* was mixed in different ratio. After treatment with leaf powder at different concentration on adult mortality against *R.dominica* infesting rice and *C.maculatus* infesting gram. Five pairs of freshly emerged adult of *R.dominica* (Fab) & *C. maculatus* (Fab) were released separately in each of experimental replication. The weight taken for each experiment was 100gm of rice and gram respectively. All experiment were setup at same temperature and relative humidity 60-70%. Six does of plant pesticides were used for such treatment & result showed quicker mortality in different percentage that depends upon dose & durations.

Keywords : *Eichhornia crassipes*, *Ipomoea carnea*, *Acorus calamus*, *Rhizopertha dominica*, *Callosobruchus maculatus*, petroleum ether extract.

INTRODUCTION

Recent year have seen a rapid advancement in the field of storage of food grains. However their is lack of common storage practice even in a given agricultural area, comprehensive principle to storage practice have to be developed, keeping a view on its ecofriendly nature and human health.

A lot of work has been done on the insect infestation in different food grains but the little or not work has been found about effectiveness of *Eichhornia*, *Ipomoea* and *Acorus* were used as pesticide. In present time Indian have

fallen serious victim from pesticidal poisoning, in India there are two law to regulate 'prevention of food alternation act 1954 and insecticidal act 1968'. It has appreciably contributed towards improving general economical condition. However there is growing awareness among planner, manufacturer and researchers about introducing newer and safer method of pest and greater pest speciality in order to keep ecological balance from pollution hazard and fulfill the concept of sustainable development.

The present work is being undertaken to examine the effectiveness of *Eichhornia*, *Ipomoea* and *Acorus* against two most common pests like *R.dominica* and *C.maculatus* and find the ideal plant products to check loss of grains through mortality of attacking pests.

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Statistical analysis was done for variability of result obtained through various parameters. Thus ratio of variation is the proportional variation in a set of two given series. The measure of proportional variation made to serve about effectiveness of plant product. This variance measured by finding the average of square of each Items of series of variable bears to mean.

After analysis of statistical, graphical and tabular approach, in graphical and tabular approach the variation. In percentage mortality considered and slightly differ from statistical approach. The mortality ratio is defined as the ratio of insects dead out of total number of insect during experiment period.

MATERIALS & METHODS

1. *Eichhornia crassipes* 2 *Ipomoea carnea* 3 *Acorus calamus* 4 *Rhizopertha dominica* 5 *Callobruchus maculatus* 6 rice 7 gram

A. Procurement of plant products, food grains and insects, leaves of plant *Eichhornia*, *Ipomoea* and *Acorus* were taken and dried in sun light then grinding into powder by mixer machine mixed form in different ratio 2E: 1I:2A and 1E: 2I:1A. After verification both grain were kept in incubator to check moisture at temp. 30 to 40°C

B. Treatment- six doses (0.5gm. 1gm.1.5gm. 2gm. 2.5gm and 3gm) of plant product treated on adult population of both pests in mixed form in different ratio.

C. Analysis- Calculation of mean of mortality percentage of pests by following formula:-

Mean =

$$\sum x \div N$$

Where $\sum x$ =sum of variations

N=no. of total variant

Now. comparative statistical analysis of total mean after treatment with mixture of leaf powder *Eichhornia*, *Ipomoea* and *Acorus* on adult mortality of *R.dominica* and *C.maculatus*.

A In ratio 2E:1I:2A

Treatment	Against <i>R.dominica</i> infesting rice(A)		Against <i>C.maculatus</i> Infesting gram (B)		Difference (M)
	Mean $\bar{X}$	Rank	Mean $\bar{X}$	Rank	
0.5gm	51.80	5	50.40	6	1.40
1.0gm	60.12	4	58.20	5	1.92
1.5gm	66.40	3	69.60	4	-3.20
2.0gm	84.60	2	74.80	3	9.80
2.5gm	95.00	1	90.33	2	4.67
3.0gm	95.00	1	93.00	1	2.00

Total mean of A = 452.92/6=75.49

Total mean of B =436.33/6=72.72

Total mean of differences (M) =
16.59/6=2.76

B In ratio 1E :2I :1A

Treatment	Against <i>R.Dominica</i> infecting rice(A)		Against <i>R.Dominica</i> infecting rice(A)		Difference (M)
	Mean $\bar{X}$	Rank	Mean $\bar{X}$	Rank	
0.5gm	22.60	6	18.00	6	4.60
1.0gm	29.80	5	30.40	5	-0.60
1.5gm	38.40	4	37.20	4	1.20
2.0gm	59.60	3	53.60	3	6.00
2.5gm	69.20	2	66.40	2	2.80
3.0gm	79.20	1	78.40	1	0.80

Total mean of A =298.80/6=49.80

Total Mean of B=284.00/6=47.33

Total Mean of Difference(M)
=14.80/6=2.47

RESULT

According to above table, the ranking of mean clearly shows highest mean shows highest mortality rate while minimum mean shows least mortality rate of insects. It is also clear that compare of rank mean shows *C.maculatus* is slightly less mortality than *R.dominica* it is also clear from above table that conc. of *Eichhornia* and *Acorus* is more effective than *Ipomoea*.

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Biochemical analysis of *Andrographis paniculata* used in diabetes in Santal Parganas, Jharkhand

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Abstract- *Andrographis paniculata* is locally known as Kalmegh or Chiraita. It is a member of family Acanthaceae. *Andrographis paniculata* is very useful for the diabetic patients in tribal locality of Deoghar and Banka district. The stems and leaves of this plant are dipped in drinking water in the night and taken in empty in the morning. Biochemical analysis of this medicinal plant has been discussed in the text in detail.

Keywords : Biochemical analysis, Amino acid, Proteins, Chiraita, Santal Pargana.

INTRODUCTION

Andrographis paniculata is locally known as 'Kalmegh' or Chiraita. It is a member of the family Acanthaceae. Local Vaidyas and Hakims of Santal Pargana used this plant for diabetic patients. The stem leaves of this plant are deeped in the night in drinking water and the extract along with drinking water is taken by the diabetic patient in the morning in empty stomach. Considerable work has been made on medicinal plants, but no work has yet been made on *Andrographis paniculata*. To fill this gap, the present investigation has been undertaken.

MATERIALS & METHODS

The extract of this plant was prepared according to methodology suggested by Ranjan and Loraya (1960)¹. Various amino acids were identified by comparing the

colour and as R/f value done on the basis of intensities which were done by visual observations recorded in 4 categories i.e. 3+, 2+, + acid and +tr. Protein analysis were determined quantitatively by disc electrophoretic method.² 10μ alcoholic extract of healthy leaves was loaded on Watman Chromatography paper No. 1 along with standard amino acids. The Chromatogram to run with solvent system containing n-butanol : acid : distilled water [4:1:1 V/V/V top layer]. They were dried, sprayed with 0.3% ninhydrin solution (300mg ninhydrin dissolved in 100 ml of n- butanol to which then 3 ml of glacial acetic acid added. The chromatography was allowed to air dry. Subsequently they were transferred to oven at 105°C for 5 minutes.

RESULTS AND DISCUSSION

Result of amino acids analysis has been given in Table 1. The seeds and another proteins (Water Soluble) were determined quantitatively by disc electrophoretic method the result of protein analysis of *Andrographis paniculata* has been in Table 2.

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Table 1. Showing data of amino acids present plant extract of *A. paniculata* collected from Inarabaran of Deoghar district.

S N.	Amino acids	Rt. values	Concentration of amino acids on visual observalia							
			Free amino acids				Bound amino acids			
			T ₁	T ₂	T ₃	Mean	T ₁	T ₂	T ₃	Mean
1.	Lysine	0.04	+	3 ⁺	3 ⁺	3 ⁺	-	-	-	-
2.	LHistidine	0.06	-	-	-	-	-	-	-	-
3.	Aspartic acid	0.09	+	+	+	+	+	+	+	+
4.	L-arginine	0.12	+	-	+	+	-	-	-	-
5.	Cysteine	0.16	3 ⁺	3 ⁺	3 ⁺	3 ⁺	+	-	+	+
6.	DL-asparagine	0.18	-	-	-	-	-	-	-	-
7.	Proline	0.21	2 ⁺	2 ⁺	2 ⁺	2 ⁺	-	-	-	-
8.	L-giutamine	0.23	-	-	-	-	-	-	-	-
9.	Serine	0.24	-	-	-	-	-	-	-	-
10.	Hydroxyproline	0.26	-	-	-	-	-	-	-	-
11.	DL-threonine	0.28	+	+	+	+	+	+	+	+
12.	Glycine	0.31	3 ⁺	3 ⁺	3 ⁺	3 ⁺	+	+	2 ⁺	+
13.	L-glutamic acid	0.33	-	-	-	-	-	-	-	-
14.	Ornithin	0.34	-	-	-	-	-	-	-	-
15.	Ditydroxy	0.37	-	-	-	-	-	-	-	-
16.	Phenylamine	0.45	2 ⁺	2 ⁺	2 ⁺	2 ⁺	-	-	-	-
17.	DL-alanine	0.49	+	+	+	+	-	-	-	-
18.	DL-methionine	0.56	-	-	-	-	-	-	-	-
19.	Amino-n-butric acid	0.61	3 ⁺	2 ⁺	3 ⁺	3 ⁺	2 ⁺	2 ⁺	2 ⁺	2 ⁺
20.	Li-ovaline Tyrosine	0.63	2 ⁺	2 ⁺	2 ⁺	2 ⁺	-	-	-	-
21.	Phenytaline	0.67	-	-	-	-	-	-	-	-
22.	DL-tryptophan	0.71	2 ⁺	+	2 ⁺	2 ⁺	+	2 ⁺	2 ⁺	2 ⁺
23.	Isoleucine	0.76	-	-	-	-	-	-	-	-
24.	Leucine	0.80	-	-	-	-	-	-	-	-

Table 2. Showing data of protein analysis of the plant extract of *Andrographis paniculata* collected from Inarabaram of Deoghar distract.

Visual band Nos.	Rp value	Visual observaliu
1.	1.3	-
2.	2.5	3 ⁺
3.	6.0	3 ⁺
4.	0.8	-
5.	3.8	-
6.	5.3	2 ⁺
7.	0.4	-
8.	3.0	-
9.	2.0	-
10.	2.3	-

Alkaloid analysis: Estimation and identification of alkaloids were done according to methods recommended by Clarke (1970)³. The result of the alkaloid analysis has been given in Table 3.

Table 3. Biochemical analysis of alkaloids of plant extract of *Andrographis paniculata* collected from Inarabaran of Deoghar district.

Alkaloids	Chromotographed Rf valuo x 100	Visual obser vath	Nature in Uv light	Reagent used for the dectectic
Cystisine	03	-	Blue	Dragen dorff
Nicotine	07	-	Absorbed	Iodoplatinate
Tomative	08	-	Invisible	Do
Morphine	14	-	Absorbed	Dragen dorff
Solanine	15	-	Invisible	- Do -
Berberine	25	-	Flourescond	- Do
Atrophine	37	-	Absorbed	Do
Quinine conine	56	-	Invisible	Do
Unknown extract	17	+	Absorbed	Dragon-dorff
	23	3 ⁺	Flurescent	Do
	48	2 ⁺	Invisible	Iodoplatinate

The plant extract of *Andrographis paniculata*. In this extract three type of alkaloid could be detected. They were 17, 23, 48 of Rf values.

Ultraviolet measurement: To confirm the presence of alkaloid, it has been measured by measuring UV spectrum of the sample dissolved in NH₂SO₄. Typical maxima values ranges from 250 to 303 nm. Alkaloids with aromatic rings in their structure may be absorbed by longer wave length. For the purpose Beekman spectrophorneter was used noted and colour of emitted light was noted on visual observation of both marker aria unknown alkaloids (Fig. 1).

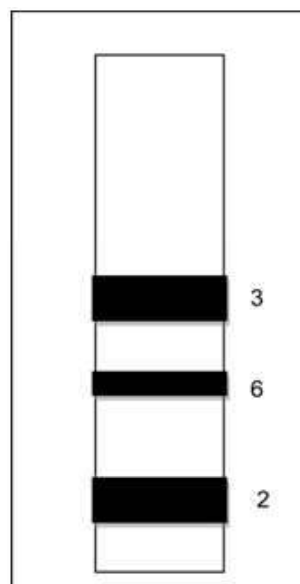


Fig. 1. Electrophoretic protein analysis of the extract of *Andrographis paniculata*

In nineteenth and twentieth century's chemical nature of natural products were developed. Currently, most of the investigation are applying their expertise to examine the biochemical nature and biosynthesis of phytochemicals to solve the stereo-chemical and mechanistic problems. However, seeing the vast number of plants relatively little has been made in this regard due to the fact that some of the principles and philosophies are often compounded by lack of the familiarity with biochemical terminology and methods. In spite of that, an even increasing of chemists are being attracted to the challenges of synthetic structural, and stereo chemical and other problems of organic chemistry.

The data presented of medicinal plant i.e. *Andrographis paniculata* Linn. which is used by diabetes patients in Deoghar district. One cup of the plant juice was used in empty stomach. One patient was sitting on the verandah of a local vaidya at Inarabaran.

When we enquired about that patient he told us that one cup of chiraita was used in empty stomach by diabetic patient, he told us that one month ago my blood sugar level was 371 but when I used chiraita one month ago my sugar level was dropped to 167 on the advice of this vaidya. I came here to show my report of blood sugar level to Vaidya.

Amino acids: In *Andrographis paniculata*, only 12 types of amine acids have been detected. These were Lysine, Aspartic acid, Larginine, Cystine proline, DL-theonine, glycine, Alarine out of there Aspartic acid, cystun, DL-theonine, glycine valine both the forms. While Lysine, Proline, DL-alanine, DL-metheorine, Trosine were recorded his free form only.

Among the 12 types of amino acid Lysine, Cystein, Glyeine and Lvaline were recorded in high potential i.e. 3⁺ level while other types of amino acids were recorded at + or 2⁺ level. Although more than 100 amino acids are known but only 20 amino acids take part is synthesis of proteins.

Protein analysis: The protein analysis of *Andrographis paniculata* were recorded the Table 2 and Fig.1. Bands were recorded at 3⁺ level while band No 6 was recorded in 2⁺ level. Having Rp value 2.5, 6.0 and 5.3.

The term protein is derived from the Greek noun protein measuring "holding first place". The term was

suggested by Berzelius (1838)⁴ and first used by G.J. Mulder in his book in 1939.⁵ Protein has very important in all the living cells. It is essential in diet for another metabolic process in living organisms.

It has varieties of roles, as proteins are of specific nature viz. nutritive, enzymatic, hormonal, the making antibodies, medicinal, genetical etc. The way all basic functions of life depend upon specific protein. Indeed we do not know life without proteins. After the works of some researchers the details of protein structure and function could be known.⁶⁻¹¹

According to them protein is a macromolecule of polypeptide chain. Therefore, polypeptide chain assume specific configuration to perform specific functions assigned to them. Protein constituents of any plant may be very helpful in tracing the phylogeny on the unknown plant. In this regard commendable works have been done by Boulter (1972)¹² MC Leester *et al.* (1983)¹³, Fair Brother (1984)¹⁴ and Neish (1996)¹⁵.

Alkaloid analysis: The chemistry of alkaloid began with the announcement of isolation of morphine form optimum in 1805 by F.W. Sertuner.

The term alkaloid was proposed by a pharmacist, C.F.W Meissner in 1819.¹⁶ Today more than five thousand alkaloids are known to occur, more than 4000 species of Angiosperms. Alkaloids are mostly Toxic substances having primary effect on central nervous system are of the organisms.

The noticeable works regarding alkaloids are of Hegnaur (1967)¹⁷, Henery (1969)¹⁸, Clarke (1970)³ and Luning (1987)¹⁹. The detailed chemistry of alkaloids could be known with the work of above biochemists. In *Andrographis paniculata* there type of alkaloid could be noticed in plant extracts. These were with Rf value 17, 23 and 48. in Uv light Rf value 17 was absorbed while Rf value 23 was with highest potencies on visual observation i.e. 3⁺ while Rf value 48 has lower potencies i.e. 2⁺. Perhaps these alkaloids lower blower blood sugar level in human beings.

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Biological control of Maize Stem Borer, (*Chilo partellus*) in Koshi region of Bihar

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Abstract- Maize stem borer damage varied from 10 to 30% in the state of Bihar. Most of the borer damage occurred when the crop was in early vegetative stage and by the time it reached tasseling, no further increase in damage was observed. The activity of adult moths started after the middle of March and continued to increase until the middle of May after which a decline in population was observed in June. This synchronizes with the infestation pattern observed for the spring maize. *Trichogramma* activity was first observed after the middle of July after which parasitism ranged from 14 to 40%. The samples collected from April to June did not indicate the presence of *Trichogramma*. Identification of *Trichogramma* species associated with maize stem borer indicated the presence of only *Trichogramma chilonis*. Preliminary studies conducted to evaluate the impact of *Trichogramma* releases showed its success in reducing the damage caused by maize stem borer.

Keywords : Maize, stem borer, *Trichogramma chilonis*, biological control, population dynamics.

INTRODUCTION

Bihar with an annual production of around 27 lakh tonnes is the second biggest maize producer in the country. It has 4.30 lakh hectares under this crop. The stretch from Purnea, Katihar and Bhagalpur to Madhepura, Saharsa, Khagaria and Samastipur, north of the Ganga and on either side of the Kosi-emerged as a corn belt. In the flood-prone areas of Kosi and parts of Seemanchal, rabi maize became the major crop. Maize cultivation has been overwhelmingly practiced by farmers across state especially in Seemanchal and Kosi region but the intra-state processing facilities

remain sick which paves the path for Bihar's dependence on other states. Productivity of maize in Bihar is 2541 kg/ha, which is far greater than the all-India average productivity of 1907 kg/ha.¹ In the absence of adequate processing facility, the huge marketable surplus of Bihar, especially in rabi season depends completely on other states for its consumption. As a result, more than 80 per cent of Bihar maize goes outside the state for processing depriving it of value addition and higher income for the people. The maize crop is a fascinating picture on the ground but policy overlook has pushed it towards a less valued agricultural output.

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Maize is an important cereal crop consumed as food, used as a fodder crop for livestock and feed for poultry, besides its use in various industrial products. It is a good source of high quality edible oil. Maize production is severely hampered by maize stem borer, *Chilo partellus*, causing losses to the tune of 15-30%.³ The damaged plants remain stunted and thus yield low fodder and grain production.

The gravid female of *C. partellus* lay eggs on the leaves and the larvae after hatching enter the stem through apical portion. Having entered the stem, the borer cannot be controlled effectively through chemicals. Systemic insecticides, though used, are not cost effective and cause environmental hazards. Biological control through the use of egg parasitoid *Trichogramma* can be very effective and economical control as it can eliminate the pest in the egg stage.^{3,4}

Because of its ease of propagation and application, *Trichogramma* is an ideal choice for inundative releases against the crop borers. But since *Trichogramma* is effective only during a short temporal window when the eggs are laid, a study of biology of pest is necessary. It is important to know the natural prevalence of *Trichogramma* in maize at different times of the year. The current studies were aimed at establishing biological control of maize stem borer through *Trichogramma*. This paper presents the results of preliminary studies needed before establishing a successful biological control programme.

MATERIALS & METHODS

Estimating the damage extent by *C. partellus*

Five villages of Madhepura of Bihar where maize is grown in big way has been selected for study. In each village, farmer's fields having maize at different growth stages were selected and data on maize stem borer infestation were recorded. Maize crop was surveyed for maize stem borer damage based on the presence of dead hearts and shot hole symptoms. The observations were recorded at a weekly interval starting from the first week of July and were continued until crop harvest.

Preliminary studies were conducted to evaluate the success of *Trichogramma chilonis* in controlling maize stem borer damage. The experiment was planned as randomized complete block design with a total of four replications. Each block consisted of 0.25 acres where

cultivar Shanker makka was sown in the first week of July. Four releases of *Trichogramma* were made starting from the first week of September. The releases were made at an interval of one week. The damage was recorded on the basis of dead hearts, a symptom of maize stem borer infestation.

RESULTS & DISCUSSION

The average bore damage varied from 10 to 30% in Kosi region. An increasing trend of damage over time was observed indicating the persistent presence of maize stem borer during the season. During August, damage was recorded which indicates the severity of borer damage. Maize stem borer leads to a severe reduction in yield. Neupane *et al.* (1985)⁵ have reported a yield reduction up to 60% in case of severe infestation. A yield reduction of 2121 Kg/ha has been reported in spring maize by Mohyuddin and Attique (1978)⁶ when 25.2% of the plants were infested while a yield reduction of 2860 kg/ha was observed in autumn when 33% plants were infested. The major loss in grain yield is due to dead-hearts and stunting of growth.

In the present findings most of the borer damage occurred when the crop was in the early vegetative stage and by the time it reached tasseling no further increase in damage was observed. This indicates the importance of emphasizing the application of biological control agent when the crop is 2-3 week old and continuing it for 4-6 weeks.

The activity of moth adults started after the middle of March and continued to increase until the middle of May after which a decline in moth population was observed in June. This synchronizes with the infestation pattern observed for the spring maize. During the months of July and August, 3-4 population peaks of moths were observed indicating the presence of overlapping moth generations during the growth period of summer maize.

The studies were initiated in July, data on *Trichogramma* prevalence was not present for that year before July. *Trichogramma* activity was first observed after the middle of July after which parasitism ranged from 14 to 44%. The samples collected from April to June did not indicate the presence of *Trichogramma* until June. Although a higher prevalence of *Trichogramma* was recorded late in the cropping season (late August to

September), no parasitoid was observed before the middle of July. This indicates that the natural populations of *Trichogramma* are not helpful in reducing the damage as they appear late where the damage is already done by *C. partellus*. Thus there is need of augmentative release earlier during the season to check the populations build up of *C. partellus*.

All the specimens collected at different locations belonged to one species i.e. *Trichogramma chilonis* Ishii. Neupane *et al.* (1985)⁵ have also reported *Trichogramma chilonis* to be the most important egg parasitoid of *C. partellus*.

Preliminary studies conducted to evaluate the impact of *Trichogramma* releases in reducing the damage caused by maize stem borer showed that the application of *T. chilonis* led to 68% decrease in damage over that of control indicating a success of *T. chilonis* against maize stem borer (Table 2).

Table 1- Incidence of maize stem borer damage during the whole crop season at different area of Kosi region

Location	Crop stage	Infestation
1	Vegetative	8.68
	Vegetative	10.5
	Vegetative	11.25
2	Tasseling	11.6
	Tasseling	12.0
3	Silking	11.8
	Silking	12.2
	Silking	12.6
4	Earing	12.0
5	Grain filling	12.8
	Grain filling	14.0
	Grain filling	12.8
	Grain filling	13.0

Table 2. Effect of *T. chilonis* application on maize stem borer damage.

Sl.no.	Treatment	Percentage damage by stem borer
1	Control	32.33
2	Treated with <i>Trichogramma</i>	10.25

So far use of systemic insecticides has been considered the only effective way of controlling maize stem borer infestation.^{7,8} Application of insecticides not only increases the cost of production but also disturbs the non target fauna of beneficial insects and microbes in the soil. *Trichogramma*, though an effective parasitoid, but present in small numbers in the nature. Augmentative releases of *T. chilonis* are recommended as a safer alternative to the use of insecticides for the management of maize stem borer.

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Benthic macrobiota in Teesta River of sub Himalayan region

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Abstract- The density of benthic macro-biota in number over weight of biomass at three sampling stations of Teesta river sub Himalayan region has been studied, it was found that the Caddis fly larvae was the most dominant component and contributed 52.41% in the macro-biota. These animals were abundant in Teesta river when benthic algae were abundant, water is well oxygenated, alkaline and contains sufficient nutrients but observed to be less abundant during high velocity of water, high river depth and higher turbid waters.

Keywords : Benthic, Macrobiota, Odonata, Teesta

INTRODUCTION

Aquatic ecosystem is one of the most productive ecosystems in the world that inhabits a large proportion of the earth's biodiversity.^{1,2} Numerous plants and animals, ranging from microscopic algae to large plants, from protozoan to mammals, exhibit a variety of adaptations which allow them to survive and grow in water.³ In the past, 'Water' the basic amenity for living organisms was pure, virgin, undisturbed, uncontaminated and basically most hospitable for living organisms but the situation is just the reverse today because progress in science and technology is also leading to pollution of environment and serious ecological imbalance which in the long run, may prove disastrous for mankind,⁴ thereby affecting its

ecological integrity to a greater extent. Macrobenthic invertebrates refer to the organisms that inhabit the bottom substrates (sediments, debris, logs, macrophytes, filamentous algae etc.) of aquatic habitats, for at least part of their life cycle. The density of aquatic macrobenthic invertebrate species and communities is controlled by a variety of environmental factors such as habitat characteristics sediment quality,⁵ sediment grain size and by biological factors such as competition and predation. Stream flow, nature of substratum and organic pollution generally regulates the species composition and dominance of different taxa in various stretches of rivers and thus, macrobenthic invertebrates constitute the most popular and commonly used group of freshwater organisms in assessing water quality. Benthic invertebrates have been favoured in environmental effects monitoring because they are sessile or limited in their range of movement and therefore

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cannot avoid pollution. They are generally abundant and can be found year round so are easily sampled. Since, many aquatic species have a life span in water of approximately a year, they provide an indication of water quality conditions over that period. Thus, benthic invertebrate monitoring data provides a link between the effects of human activities on the physical and chemical properties of water and aquatic ecosystem health.⁶

MATERIALS AND METHODS

The samples of macro-benthic animals were collected on monthly basis by "the kick net method" in which 1 m 2 stream bed areas was enclosed with net fabricated from fine mesh, soft and flexible monofilament material. The entire fauna collected and preserved in 5-10% formalin for further detailed analysis.

RESULTS AND DISCUSSION

Several investigations have been made on the nature, density and ecology of the macro-benthic animals of Indian rivers and streams. Singh and Nautiyal (1990)⁷ studied benthic fauna of river Ganga. In the whole stretch of river, 30 taxa of macro invertebrates dominated by Ephemeroptera followed by Diptera and Trichoptera. Mohan and Bisht (1991)⁸ recorded species of insects belonging to 37 families from River Bhagirathi and Bhilangana. Sehgal (1991)⁹ recorded 57 genera of insects from 11 rivers of North-Western Himalaya. Joshi (1991)¹⁰ observed 50 genera of insects from Sherkhad stream in Himachal Pradesh. Bhatt and Pathak (1992)¹¹ recorded 68 genera of insects from various rivers of Kumaon region and 37 taxa from Teesta river.

Table 2 :Average annual percentage contribution of various macro-benthic animals at three sampling sites of River Teesta.

Groups		Site 1	Site 2	Site 3
Ephemeroptera				
	<i>Epeorus</i>	26.47	30.77	33.57
	<i>Heptagenia</i>	11.34	10.99	7.00
	<i>Rhithrogenia</i>	15.97	7.14	12.59
	<i>Caenis</i>	4.20	6.59	2.80
	<i>Baetis</i>	16.39	14.84	11.19
	<i>Ephemerella</i>	7.98	13.19	7.00
	<i>Echdyonurus</i>	8.40	1.54	7.68

	<i>Iron</i>	5.88	3.85	9.08
	<i>Leptophlebia</i>	2.10	1.09	-
	<i>Ameletus</i>	1.27	-	8.40
	<i>Paramelatus</i>	-	-	0.69
Odonata				
	<i>Argus</i>	3.92	4.26	11.63
	<i>Gomphus</i>	45.10	25.53	37.21
	<i>Octogomphus</i>	15.68	8.51	-
	<i>Agriion</i>	15.69	21.28	32.56
	<i>Ophiogomphus</i>	-	4.25	13.95
	<i>Ischnura</i>	19.61	36.17	4.65
Plecoptera				
	<i>Perla</i>	32.76	28.12	21.43
	<i>Chloroperla</i>	27.59	18.75	28.57
	<i>Nemoura</i>	36.21	50.00	45.24
	<i>Leuctra</i>	3.44	3.13	4.76
Hemiptera				
	<i>Micronecta</i>	100.00	81.82	100.00
	<i>Corixa</i>	-	18.18	-
Coleoptera				
	<i>Psephenus</i>	72.84	83.33	85.56
	<i>Elmis</i>	17.28	11.54	6.67
	<i>Ectopria</i>	8.64	-	3.33
	<i>Eubrianex</i>	-	1.28	4.4
	<i>Hydrena</i>	1.24	2.57	-
	<i>Dysticus</i>	-	1.28	-
Trichoptera				
	<i>Hydropsyche</i>	61.52	63.86	62.80
	<i>Rhyacophylla</i>	21.77	18.52	27.68
	<i>Phyllopotamus</i>	15.70	17.32	9.52
	<i>Mystacids</i>	0.38	0.70	-
	<i>Ithytrychia</i>	0.63	-	-
Diptera				
	<i>Chironomus</i>	40.58	47.34	63.53
	<i>Atherix</i>	14.50	17.55	14.12
	<i>Dixa</i>	1.93	1.06	-
	<i>Tabanus</i>	16.91	9.04	3.52
	<i>Chaoborus</i>	14.01	8.52	8.23
	<i>Cullicoides</i>	1.45	-	-
	<i>Simulium</i>	8.70	14.36	10.59
	<i>Pentaneura</i>	1.44	2.13	-
	<i>Psychoda</i>	0.48	-	-

During the present investigations, a total of 48 genera of macro-benthic animals were recorded. Sehgal (1990)¹² reported univoltine type of life cycle in may flies and stone flies but caddis larvae exhibit multivoltine type of life in upper reaches of River Beas in Himachal Pradesh. During the present studies in Teesta river, may flies exhibited peak in November, February and May of which the peak in February was strongest due to the addition of new arrivals in the population. In the case of caddis fly, clear peak was

seen in their numerical density during November, February and May in which newly entrants to population were prominently observed in November and May along with larger sized cohorts. Among the midge larvae, *Chironomus* had its peak density in September just at the end of Southwest monsoon when newly hatched (almost transparent) *Chironomus* were collected not only in macrobenthic samples but also in zooplankton and epiphytic algae samples. The second peak was observed in January at Station 3 and third peak in May. This indicates that other than may flies which appear to lead univoltine type of life, caddis fly and midge larvae indulge in multivoltinism in Gaula river.

The great variability of a stony stream, bed which offers numerous different microinhabitat is the most important factor which determines the distribution and abundance of macrobenthic animals. The substratum acts directly as a medium for their existence and indirectly as modified of their microenvironment (Minshall, 1984)¹³. There may be many types of these habitats within even a small portion of stream. They are partly determined by the speed of the current but often varies considerably according to the kind, size and shapes of the stones, over or round which the water flows. In India, Hora (1936)¹⁴ conducted premier investigations on benthic invertebrates and considered substratum as an important factor in the ecology of torrential fauna.

Sehgal (1988, 1991)^{15,9} dealt with this aspect in detail and provided authentic explanations Negi and Singh (1990)¹⁶ dealt with substratum in relation to bottom fauna of river Alaknanda in Garhwal Himalaya. In Gaula river, substratum near Herakhan comprised of big and small stones and boulders; at Station 2 near proposed Jamrani Dam site has small and big stones and sandy; at Station 3 near Ranibagh, the site is mostly of sand, small pebbles and small stones. The average annual numerical density at Station I was estimated at 80.27 ind./m², at Station 2, 67.5 ind./m² and at Station 3, 41.94 ind./m². This indicates that macro-benthic fauna in Teesta river prefer river bed substratum comprised of small and big stones and boulders where average density of fauna was almost double when compared with sandy habitats at Station 3. A good number of *Epeorus* (15 no./ m²) were collected by washing clusters of filamentous green algae *Zygnema* and *Tribonema* in September 1993 at Station 1 and in October 1993 at Station

2. The nymphs of *Rhithrogenia* (17 ind./m²) and *Baetis* (13 ind./m²) were collected from small crevices of submerged stones with the help of a dropper. But midge larvae particularly *Chironomus* in their early larval forms were found accumulated (26 ind./m²). at Station I and 37 ind./m² at Station 2) along with lesser density of *Atherix* (10 ind./m² at Station 2) under a thin algal biofilm near the bank of river in the month of September 1993 when water velocity at these two sampling stations was considerable. The omnivore *Hydropsyche*, *Rhyacophila* and *Phyllopotamus* were mostly collected by kicking stones as they were mostly available in between these microhabitats and only few of them were attached to stones.

Water penny *Psephenus* and *Elmis* were collected by washing stones as they remain attached to them. In Indian waters, Khan and Tandon in their study on reappearance of benthic invertebrates in River Beas observed that on an average period taken by the upland river fauna to reappear after complete wash off is about 63 hours during March compared to 138 hours in October. In Gaula river may flies were least available in August and September 1992 may be due to thus wiping out completely. In the first week of September, 1992, a big part of landslides containing large and heavy boulders blocked Teesta river between sampling stations, II and III. This persisted for about a week when Irrigation Department Engineers blasted off the blockade.

When sampling was done at Stations II and III after weather conditions returned to normal, very low density of benthic invertebrates was recorded. But a good number of *Epeorus* (15 units/m²) were observed. After this main physical disturbance of flash floods which increased the depth of the river, water velocity and finally turbidity subsequently brought down numerical density and wet biomass of benthic animals. The settlement of may flies started and attained their first peak (24 units/m²) in November. Their density came down in December and January but again got colonised in good quantity (48 units/m²) at sampling Station I, Again a minor peak (11 units/m²) in may fly density was observed in May. Almost the same trend was observed at other two sampling stations. It appears that caddis larvae are farquicker in colonising process than others. At sampling station I, only 6 units/m² of caddis fly larvae recorded rising to 44 units/m² in

October and 101 units/m² in November which was the numerical maxima. Like many flies, their density came down in December and January but regrouped again in February (88 units/m²). Subsequently their density came down in March and April and due to increase in *Rhyacophila* density established their third peak (61 units/m²) in May. Midge larvae (Diptera) showed their maximum density in September 1993 at Station I (45 units/m²) and Station II (69 units/m²) and in January 1993 at Station III (20 units/m²). It is surprising that average annual number of midge larvae came down from 11.5 units/m² at Station I to 10.4 units/m² at Station II to 4.7 units/m² at Station III. Few studies have established correlation between enhanced ionic concentrations and benthic animals and their biomass. In Scottish highland streams, Egglisshaw and Morgan (1965)¹⁷ fixed certain limits in values of alkalinity. They concluded that low benthic densities occurred in streams with alkalinity less than 20 mg whereas benthic invertebrates density was quite high in streams with alkalinities higher than 20 mg. Feldman and Connor (1992)¹⁸ observed a threshold of alkalinity at 50 mg. They observed that alkalinities below 50 mg may create physiological problems in some genera especially *Ephemeroptera* and *Plecoptera*. Perhaps these groups are more prone to changes in temperature and also to dissolved oxygen.

Krueger and Waters (1983)¹⁹ suggested that secondary production was greatly affected by chemical nature of water. These studies suggested that macro benthic invertebrates density may increase directly through elevated periphyton standing crops or productivity. Other studies have demonstrated positive relationship between alkalinity and macroinvertebrate density. Krueger and Waters (1983)¹⁹ found significant correlation between drift density and alkalinity in 13 Alaskan streams. Krueger and Waters (1983)¹⁹ also found a positive relationship between alkalinity and macro-invertebrate biomass in five Minnesota streams. Hershey *et al.* (1993)²⁰ found reduced drift from enriched sections of streams. Koetsier *et al.* (1996)²¹ observed that macro invertebrates biomass showed a significant positive relationship with alkalinity.

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Effect of fenitrothion pyrethroids on the respiratory rate of Singhi, *Heteropneustes fossilis* (Bloch)

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Abstract- While using several insecticides to control insect pests especially hemipterans in the agricultural field (Paddy), it has been found many pests become resistant to the common pesticides - deltamethrin (a pyrethroid) normally used to treat the crops for making them insect free. However, the pond also provides a good shelter to the bottom dwelling fishes like Singhi, *Heteropneustes fossilis* (Bloch) which are concomitantly exposed to the pesticides being used for insect control.

In the present article the effect of a substitute pyrethroid-fenitrothion in place of deltamethrin on the respiratory rate of *H. fossilis* has been observed as because many of the farmers are trying this insecticide as safe to control the insect pests. Befittingly, the researchers are supposed to investigate the impact of safer variant on the coexisting fauna at like fishes. The effect of this pyrethroid on the respiratory function of the selected fish has been observed to be less stringent and lethal because its LC_{50} & LD_{50} values are relatively low. In the normal fishes (control) per minute opercular beat as index of respiratory function has been found to be 30 while fenitrothion treated fishes at 10%, 20% & 30% concentration (weight/volume in distilled water) for 15 days duration exhibited the increased opercular beat to 34, 39 & 52 per minute respectively. Higher concentration of fenitrothion could push the opercular beat to 52 per minute only as compared to 68 in case of Deltamethrin treatment. Gradual increasing trend by 20-30% and sharp peak of opercular beat have also been recorded in 30 and 45 days of exposure leading to high rate of casualty (in 30% concentration and 45 days of exposure).

Keywords : fenitrothion, pyrethroid, low toxicity, LC_{50} value-41.66 & LD_{50} value-8.33, less respiratory stress, *H. fossilis*

INTRODUCTION

The respiratory function in fish is morphologically the function of pulsatory alternative opening and closing of mouth with related opercular movement. Hence, the rate of opercular beat may be taken as the quantitative parameter to evaluate the respiratory function in both

normal (control) and insecticides exposed (treated) in such cases. Recording the rate of opening and closing of mouth in conjunction with the per unit time opercular beat, although is preliminary protocol to evaluate the respiratory function, yet it is sometime misreading and unreliable as many of the fishes may stream in water current through nostrils and side clefts of oral aperture (cartilaginous fishes) to import the dissolved oxygen without opening and closing of mouth.¹⁻³

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The physiological profile of consuming dissolved oxygen at cellular level during metabolic process linked with rate and efficiency of transportation of oxyhaemoglobin through RBC in the bold of fishes. Significantly depend on the rate of opercular beat, hence both these parameters directly impact each other. Therefore, in the present investigation ample focus has been given on the accurate counting of opercular beat per unit time in the test fishes both in control and treated conditions.⁴⁻⁶

MATERIAL & METHOD

35 adult fishes (*Heteropneustes fossilis*) irrespective of sex consideration weighing 200-225 gm were collected in the control tank in the lab from the local market of Madhepura having same shape and size. Keeping 5 fishes in the reserve tank as control 30 fishes were distributed in the group of 5 which were transferred to six different tanks of experimental treatment. Three sets of tanks (containing three in each) with 10%, 20% and 30% of fenitrothion concentration marked for 15 days, 30 days & 45 days duration of exposure were installed for experimental observation with respect to altered rate of opercular movement in different concentration and exposure gradient.

The standard rate of opercular beat (OB/OM) was recorded from the fishes of controlled tank and the changed respiratory behavior if any in the treated fishes kept in three sets of tanks were recorded separately in the observation table.

Multiple observations (at least five) for counting of opercular beat per unit time with the help of stop watch and electronic alarm counter (counting machine) were taken to calculate the mean value of opercular beat in the experimental set up and correlate it with the overall respiratory function.

RESULT & DISCUSSION

The present study exposure of fish to sub lethal concentration of fenitrothion in the range of 10%, 20% & 30% with LC50 value- 41.66 & LD50 value- 8.33 for 15, 30 & 45 days caused gradual but significant alterations in opercular beat being the index of respiratory function of the fish, *Heteropneustes fossilis*. The alterations in the rate of per minute opercular beat were found to be asphyxiating for the fishes in tank no. 3 at 30% concentration of fenitrothion with 45 days exposure. The effect of deltamethrin on the respiratory function of this fish as observed in previous investigation⁷ was more stringent as the opercular beat got elevated to 68 beats per minute in 10% concentration only for 15 days duration. Therefore, the LC50 & Ld50 values of the previous insecticides deltamethrin were also high resulting into high degree of casuality of the fishes in the agricultural crop field.

In order to protect the life of the Singhi fish which is a delicacy food of the area and also to keep the crop free from the attack, lower concentration with mild dose spraying of fenitrothion in recommended for the health and benefit of agricultural crop and all components of the ecosystem.

Table 1.- Gradient of opercular beat per unit time as indicator of respiratory functions in *H.fossilis* treated with 3 different concentration of fenitrothions for 15 days duration

No. of Observations of respiratory function in terms of opercular beat/minute	Control	Altered rate of opercular beat in different concentration of fenitrothion treated fishes			LC50	LD50
		10%	20%	30%		
1	30	35	41	58	41.66	8.33
2	30	33	38	45		
3	30	34	39	62		
4	30	35	38	51		
5	30	33	39	44		
Mean value	30	34	39	52		

Table 2.- Gradient of opercular beat per unit time as indicator of respiratory functions in *H.fossilis* treated with 3 different concentration of fenitrothions for 30 days duration

No. of Observations of respiratory function in terms of opercular beat/minute	Control	Altered rate of opercular beat in different concentration of fenitrothion treated fishes			LC50	LD50
		10%	20%	30%		
1	30	39	45	59	46.66	9.33
2	30	38	44	69		
3	30	36	42	61		
4	30	38	43	59		
5	30	39	46	48		
Mean value	30	38	44	58		

Table 3.- Gradient of opercular beat per unit time as indicator of respiratory functions in *H.fossilis* treated with 3 different concentration of fenitrothions for 45 days duration

No. of Observations of respiratory function in terms of opercular beat/minute	Control	Altered rate of opercular beat in different concentration of fenitrothion treated fishes			LC50	LD50
		10%	20%	30%		
1	30	41	47	60	50.00	10.00
2	30	44	48	59		
3	30	41	45	61		
4	30	43	49	64		
5	30	41	51	56		
Mean value	30	42	48	60		

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Seasonal variation of physico-chemical status of Kanjha Kothi Lake of Purnia

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Abstract- The study of physico-chemical parameters of Kanjha Kothi Lake were conducted to study the present of water for pond fish culture. Water samples were collected from the ponds and analyzed using standard laboratory methods and procedures. The results showed variation in the observed parameters at the different sampling stations. The physico-chemical parameters i.e. temperature, pH, dissolved oxygen, total alkalinity, free carbon dioxide, transparency, hardness, total dissolved solids, Conductivity and biological oxygen demand has been studied and correlated with ideal condition for fish culture.

Keywords: Physico-chemical, Kanjha Kothi Lake, Purnea, Fish culture

INTRODUCTION

Water is the living media for fish and its quality is one of the most over looked aspect of pond management until it affects fish production. Water quality generally means the component of water which must be present for optimum growth of aquatic organisms.^{1,2} Water quality is made up of physical, chemical and biological factors which influence the use of water for fish culture purposes. These factors include temperature, pH, dissolved oxygen, total alkalinity, free carbon dioxide, transparency, hardness, total dissolved solids, conductivity and biological oxygen demand. The biological oxygen demand and chemical oxygen demand indicate the pollution level of a given water body.¹ Productivity depends on the physico-chemical

characteristics of the water body.³ In recent years, the inland waters and the terrestrial life in the state have been subjected to alteration ecologically. This is partly due to the human activities, population growth, oil exploitation and exploration which resulted in the pollution of the environment. The components of the pollution contribute to greater oxygen demand and nutrient loading of the water bodies, promoting toxic algal blooms and leading to destabilized aquatic ecosystem.⁴ Water of such poor quality may be acidic, rich in nutrients and organic matter, high in suspended solids or polluted with industrial or agricultural chemicals. Such deterioration in water quality can result in fish being stressed and vulnerable to disease.⁵ Fish ponds in the State play a vital role in fisheries to supply the much needed animal protein to the people inhabiting such areas. It is therefore important to know about water

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quality parameters and their management which have influence on growth and survival of aquatic organisms. The purpose of the present investigation was to provide data on water chemistry and biological characteristics of fish ponds in selected freshwater areas of the State. Limnological investigations on water bodies were generally analyzed to determine the pollution level of the water bodies time to time. The abiotic and biotic factors of the water influence the quality and quantity of aquatic life surviving there. The role of water in nature is unique not only for human; but, also for the numerous organisms living in the water. The physical and chemical properties of fresh water bodies are characterized by the climatic, geochemical, geo morphological and pollution condition. In order to utilize fresh water bodies successfully for fish production, it is very important to study the physico-chemical factors influencing the biological productivity in the water bodies.² The quality of aquatic life surviving in the pond is totally dependent on the water quality of the pond. In the recent years several studies have been made in this field⁶ not much information is available on physico- chemical and biological parameters of the present water bodies. The present study will give an idea to increase the conducive condition to fish culture and increase the fish production.

MATERIALS & METHODS

Three points of Kanjha Kothi Lake were selected for the study. Monthly sampling was done from June 2016 to May 2017 in all the three ponds for finding out the various abiotic (temperature, transparency, pH, dissolve oxygen, free CO₂, total alkalinity, BOD) and biotic parameters. The physico-chemical analyses of the water samples were done according to standard methods.⁷ For estimation of planktonic population, samples were collected with the aid of a mug of one litre capacity. 50 litres of water was collected through the plankton net made of bolting silk no. 25 (0.064 mm mesh size) to obtain the plankton sample and was preserved by adding 2- 4% of formalin for further study. The data collected was grouped in three categories i.e. summer (February-May), Rainy (June-September) and winter (October-January) for comparison of seasonal variations.

RESULTS AND DISCUSSION

The physico-chemical analysis made in various seasons in these ponds has been shown in the Table 1.

Temperature

Temperature plays an important role in aquatic environment and considered as an important factor in controlling the functioning of aquatic ecosystem.⁸⁻¹⁰ In the present study seasonal variability of atmospheric and water temperature have been observed. The air temperature varied from 25°C to 31°C whereas the water temperature range 23.0°C to 27.0°C. It was maximum during summer comparatively less during monsoon and minimum during winter. Similar results have been observed by other workers.¹¹ There was no variation in temperature in all the three points. The temperature ranges indicate the suitability for fish culture¹² even though the minimum temperature recorded in the month of December (16°C) is also within the permissible limit of the fish culture. Due to large size of the water bodies there are no significant changes in the water temperature. In the peak summer season there was very minor variation in pond 1 in comparison to Pond 2 & 3, may be due cemented embankment and comparatively smaller in size.

pH

pH is a limiting factor and works as an index of general environmental condition. The pH value of the pond showed alkaline trend. The maximum pH value were in the month of April i.e. 8.5 and minimum in the month of October i.e. 7.85. It is evident from the data that the pH declines during the rainy season and increases during summer. pH status in the aquatic environment on important chemical parameters which predicts about the suitability for the fish culture. The alkaline pH is suitable for fish culture.¹² The pH was more in where there is washing of the cloths and less where the cattle waste are disposed, but overall the pH was same and was little alkaline which is most suitable for the fish culture.

Transparency

Water transparency is an important factor that controls the energy relationship at different trophic levels. The results of transparency ranged between 28 cm to 36 cm, during the study period. It was low during the summer and higher during the winter season. The transparency was lower in the summer season due to high planktonic population, while it was low in the rainy season because of increase in the suspended matter brought in through surface run off. The maximum transparency was recorded

in winter season attributed to the sedimentation of suspended matter.¹³⁻¹⁶ Transparency plays vital role in nutrient transformation and also gives an indication of productivity of the water. The availability of the plankton influences the transparency of the water. The water was more transparent in the rainy season in comparison to the winter and summer season may be due to poor diluted plankton in the rainy season.

Dissolve oxygen

Oxygen content is important for direct needs of many organisms and affects the solubility of many nutrients and therefore the periodicity of aquatic ecosystem.⁸ Jhingran (1982)¹² stated that the oxygen contents in tropical water would be low considering their high temperature. The results of the present study showed that highest peak value of dissolved oxygen was recorded during the winter season i.e. 9.5 ppm least in rainy season i.e. 8.5 ppm. The minimum dissolved oxygen has been noticed in summer may be due to excessive temperature which reduces the solubility of oxygen. The higher temperature also increases the decomposition rate and the lowers the oxygen Results of the present study are similar to other workers.^{17,18}

Carbon dioxide

The normal water receive carbon dioxide from various sources i.e.

- (1) The atmosphere.
- (2) Respiration of plants and animals.
- (3) Bacterial decomposition of organic matter.

Inflowing ground water. The carbon dioxide bears a co-relation with pH. The increase in carbon dioxide decreases pH (acidic). The CO₂ varied from 16 to 20 ppm. Maximum free CO₂ in was observed in summer season and minimum in the rainy season. The free carbon dioxide concentration depends on the respiration of organism (plants and animals) and photosynthesis rate. In case of more photosynthesis more carbon dioxide will be utilized.

Alkalinity

Alkalinity is a function of bicarbonate and carbonates. These salts get hydrolyzed in solution and produced hydroxyl ion. It is also used as a measure of productivity of water.^{12,19} Natural water bodies in tropics usually show wide range of fluctuations in their total alkalinity value depending upon the geography and season. In the present study the total alkalinity ranged between 190 ppm to 224

ppm. It is gradually decreased from July to September and then increased in the month of October. Seasonally highest value was recorded during rainy and lowest during the summer season. Increases in total alkalinity during rainy season were due to input of water and dissolution of calcium carbonate ion in the water column.²⁰ The degradation of plants and other organism and organic waste might also be one of the reason for the increase in carbonate and bicarbonate thereby the alkalinity.^{21,22}

Hardness

Hardness in water is due to salts of Ca⁺⁺ and Mg⁺⁺ mainly in the form of carbonates and sulphates.²³ In the present study the total hardness of water ranged from 90-120 ppm seasonally, highest value was recorded during summer and lowest during the rainy season. Similar observations were found by various workers^{19,24-27} other workers has also found¹⁹ found that higher value in summer and lower in winter season. They attributed is to decreases in water volume and increases in rate of evaporation at high temperature. The water can be categorized according to degree of hardness as soft (0-75 mg/l) moderately (75-150 mg/l) hard, hard (150-300 mg/l) and above 300 mg/l as very hard. On the basis of the observation, the water of the present pond appears to be hard.

Electrical conductivity

Electrical conductivity of the water depends on the nature and concentration of salts in high ionic concentration, pollution status, trophic levels, some domestic effluents and other organic matter in water.²⁸ The range of electrical conductivity in the present study was between 2.4 mScm-1 to 2.9 mScm-1. The values of electrical conductivity showed marked seasonal variation being maximum during rainy and minimum during winter season. Similar results were observed by various workers.^{18,19}

Total dissolve solids

Water is a universal solvent and has a large number of salts dissolved in it which largely govern the physico-chemical properties. The maximum value of total dissolved solids was recorded in rainy season 172 ppm and minimum were recorded in winter season 154 ppm. The high value of TDS during rainy may be due to addition of domestic waste water, garbage and sewage etc. in the natural surface water body.

Biological oxygen demand

BOD is dissolved oxygen required by microorganism for aerobic decomposition of organic matter present in water. Other worker considered BOD as an important parameter in aquatic ecosystem to establish the status of pollution.²¹ The observation of present study showed that highest value of BOD value during rainy season 3.2ppm and lowers during winter season 2.21ppm. There was variation in pond 1 and pond 2 &3 may be due to the waste matter flowing in the pond from the catchment. Seasonally, the BOD was highest during late summer/early rainy season. High BOD during late summer/early rainy season

may be due to the presence of several microbes in water bodies which accelerate their metabolic activities with the increase in concentration of organic matter in the form of municipal and domestic waste pouring into the pond with run off. It is reported that the higher values of BOD during rainy was also due to input of organic wastes and enhanced bacterial activity. High temperatures do play an important role by increasing rate of oxidation. The BOD of unpolluted water is less than 1.00 ppm moderately polluted water 2.00-9.00 ppm while heavily polluted water have BOD more than 10.00 ppm. The BOD in different season in the present study indicates pond as moderately polluted.

Table 1: Details of the Physico-chemical and biological parameters of the ponds

Parameters	Site 1			Site 2			Site 3		
	Summer	Rainy	Winter	Summer	Rainy	Winter	Summer	Rainy	Winter
Air Temperature (°C)	31.47 ±2.0	27.3 ±2.1	25.2 ±2.1	31.47 ±2.0	27.3 ±2.1	25.2 ±2.1	31.47 ±2.0	27.3 ±2.1	25.2 ±2.1
Water Temperature (°C)	27.4 ±0.8	24.4 ±1.5	22.95 ±1.5	27.4 ±0.8	24.4 ±1.5	22.95 ±1.5	27.4 ±0.8	24.4 ±1.5	22.95 ±1.5
pH	8.5 ±0.4	8.0 ±0.3	7.85 ±0.25	8.5 ±0.4	8.0 ±0.3	7.85 ±0.25	8.5 ±0.4	8.0 ±0.3	7.85 ±0.25
Dissolved oxygen (ppm)	8.7 ±0.4	9.6 ±0.3	9.5 ±0.2	8.7 ±0.4	9.6 ±0.3	9.5 ±0.2	8.7 ±0.4	9.6 ±0.3	9.5 ±0.2
Free Carbon dioxide(ppm)	16 ±2.0	18 ±3	20 ±3	16 ±2.0	18 ±3	20 ±3	16 ±2.0	18 ±3	20 ±3
Alkalinity(ppm)	190 ±5	220 ±3	202 ±3	190 ±5	220 ±3	202 ±3	190 ±5	220 ±3	202 ±3
Secchi disc transparency (cm)	28.5 ±1.5	36 ±3.0	29.5 ±2.0	28.5 ±1.5	36 ±3.0	29.5 ±2.0	28.5 ±1.5	36 ±3.0	29.5 ±2.0
Hardness(ppm)	118 ±3	90 ±5	103 ±2.5	118 ±3	90 ±5	103 ±2.5	118 ±3	90 ±5	103 ±2.5
Conductivity	2.64 ±0.5	2.83 ±0.5	2.42 ±0.6	2.64 ±0.5	2.83 ±0.5	2.42 ±0.6	2.64 ±0.5	2.83 ±0.5	2.42 ±0.6
Total Dissolved solids(ppm)	165 ±10	172 ±8.0	154 ±6.0	165 ±10	172 ±8.0	154 ±6.0	165 ±10	172 ±8.0	154 ±6.0
Biological oxygen demand (ppm)	2.81 ±0.4	2.82 ±0.5	2.21 ±0.6	2.81 ±0.4	2.82 ±0.5	2.21 ±0.6	2.81 ±0.4	2.82 ±0.5	2.21 ±0.6

CONCLUSION

The observation of the different parameters revealed that the physico-chemical of all the Kanjha Kothi Lake is within the permissible limit for fish culture except the concentration of plankton. It may be due to over stocking or more organisms feeding on plankton. It may be recommended that the stocking of the pond should be done on the basis of the natural productivity of the water or supplementary feeding (if possible) to have better

production. Our analysis for BOD of the ponds falls under moderately polluted category which should be solved in future for better productivity.

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Studies on the morphology of the malpighian tubules of *Platycorynus peregrinus*

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Abstract- The present study was undertaken to know the morphology of the malpighian tubules *Platycorynus peregrinus*. This is a common AK-Chrysomelid beetle, commonly found on the Aseolpidase. This beetle is very common in and around Patna, Gaya, Hazaribagh, Nalanda and other districts of Bihar and Jharkhand on the banks of river during the month of July to November.

Keywords : *Platycorynus peregrinus*, malpighian tubules, Aseolpidase

INTRODUCTION

In dissected *P. peregrinus*, the malpighian tubules become well marked almost filling the whole body cavity. These are very long convoluted structures coiling around almost of the organ.¹⁻⁵ They are richly supplied with trachea and tracheols. These tracheols branches bind the malpighian tubules to adjacent organ such as foregut, midgut, upper portion of the hindgut and reproductive organ.⁶⁻⁹

MATERIALS AND METHOD

The adult beetles were collected from the field. For the study of malpighian tubules of *P. peregrinus*, well fed insects were taken from the cages and kept on dissecting tray for dissection.

RESULT AND DISCUSSION

In dissected *P. peregrinus* the malpighian tubules become well marked. *Platycorynus peregrinus* has six cryptonephric malpighian tubules arising at the junction of midgut and hindgut. The point of the malpighian tubules are situated at the pylorus i.e. malpighian tubules open into the pylorus or the anterior part of the hindgut.¹⁰

All malpighian tubules are extremely long, slender, unbranched and highly convoluted and the same size throughout their length. They first move upwards, hugely supplied with trachea upto the length of the midgut and then turn backwards and surrounds every organs.¹¹

All six malpighian tubules unite together at their distal ends which come close to each other forming a small swelling which is closely applied to the epithelium of the colon and this is known as cryptonephridial condition, which is a device for conservation of water in terrestrial insects.

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The cells of malpighian tubules are responsible for keeping the alimentary canal and other organs like reproductive organs also ensuring the maximum contact with the blood.

Though there is no histological difference between the proximal and distal portion of the malpighian tubules but the tubules just at the point of fusion show that the epithelial cells are more elongated and lumen is obliterated.

The cross section of the tubules show the presence of a single layer of 9-11 cuboidal epithelial cells having a granular cytoplasm and the nucleus move towards the apical zone rather than to the basal zone. The brush border is present.

On the above observations it can be concluded as the malpighian tubules of *Platycorynus peregrinus* has a very important role in the body of the insects.

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Bioaccumulation of heavy metal by water hyacinth

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Abstract : Aquatic plants are well known accumulator of heavy metals from wetland. The objective of this study is to evaluate the capacity of *Eichhornia crassipes* to accumulate lead from municipal waste water. Young plant of *Eichhornia crassipes* of equal size were grown in municipal waste water for 20 days. The plant in the experiment removed 50% lead. Removal of metal from waste water was fast in the first 10 days. Bio-concentration factor (BCF) was more than 1000. *Eichhornia* was found to be suitable plant for removal of heavy metal from waste water.

Key words: BCF, *Eichhornia crassipes*, bioaccumulation

INTRODUCTION

Heavy metals are metallic chemical elements with a high atomic weight and density much greater than water. They are highly toxic and cause adverse effects even at very low concentrations e.g. mercury (Hg), cadmium (Cd), arsenic (As), chromium (Cr), thallium (Tl), and lead (Pb). They are added to the aquatic system, either naturally by slow leaching from soil/rock to water or through anthropogenic sources.¹⁻⁵ These heavy metals are not biodegradable; hence there is a need to develop such remediation technique, which should be efficient, economical and rapidly deployable in a wide range of physical settings.⁶

Wetland treatment technology offers a comparative advantage over conventional, mechanized treatment systems as the level of self-sufficiency, ecological balance,

and economic viability is greater. Phytoremediation technology consists of mitigating pollutant concentrations in contaminated water bodies with naturally occurring or genetically engineered plants that have ability to accumulate, degrade or eliminate metals, pesticides, solvents, explosives, crude oil, and its derivatives *etc.*^{7,8}

MATERIAL & METHODS

To assess the heavy metal absorption capacity of *Eichhornia*, test was conducted in three cylindrical tanks of 1000 L capacity with 80cm radius and 50cm depth. In each tank 20cm deep 400 L municipal waste water was added. Each tank was seeded with 4kg fresh and young *Eichhornia* plants which were taken from an unpolluted pond. 1 tank was taken as control. Plants were thoroughly washed with clean water before seeding. In waste water Lead was analysed in initial, 5th, 10th, 15th and 20th day. Metal concentration of each plant was analysed on initial and final stage of experiment.

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CHEMICAL ANALYSIS

50ml of water sample at each level (in initial, 5th, 10th, 15th and 20th day) were digested in 2M HNO₃ at 95°C for 2 hours and made upto 100ml in a volumetric flask with distilled water. Digested samples analysed for metal using atomic absorption spectrophotometer.

The plant samples were separated into root and leaves to determine the accumulation trend from water to the roots and to the leaves. They were each dried in an oven at 60°C till well dried. The dried samples were ground before digestion. 500 mg of dried weight of each fraction were digested with 10 mL of HClO₄ and HNO₃ mixture (1:3) at about 80°C for 4 hours. The resulting cleared coloured solutions were made up to the mark in a 25 ml volumetric flask with the distilled water. All the reagents that were used were of analytical grade and all the reaction vessels were treated well to avoid external contributions of the metals. Sample blanks were analysed to correct the possible external contributions while replicate samples were also evaluated and all the analyses were done in triplicate to ensure reproducibility of the results. The digested samples were analysed for lead by atomic absorption spectrophotometer.

In the present investigation, the bioconcentration factor (BCF) of heavy metal (Lead) within *Eichhornia crassipes* was calculated by following formula:

$$BCF = \frac{\text{Trace element conc. in harvested plant } \left(\frac{mg}{kg}\right)}{\text{Initial conc. of element in water } \left(\frac{mg}{l}\right)}$$

RESULT

The final concentration of lead in waste water treated with *Eichhorniacrassipes* was lower than concentration of lead in control tank. The bio-concentration factor (BCF) in the present study is given in Table.

Table. 1

Metal	BCF in Root	BCF in Leaves	BCF in Plant
Pb	7625.54	2116.67	4866.15

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Scattered distribution & low abundance of worm snakes vis-a-vis blind snakes of superfamily Typhlopoidea of Madhepura, Bihar, India

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Abstract : Common worm snakes also known as blind snakes are members of several non-venomous snakes characterized by degenerate eyes that lie beneath opaque head scales. As regards distribution, they are fossorial or burrowing and are regularly associated with termite's and ant's nest (termitorium and antorium respectively) as symbionts. Different species of blind snakes belong to the families Anomalepidae, Leptotyphlopidae and Typhlopidae falling under the superfamily Typhlopoidea. However the populations of these worm snakes are very small in the terrestrial tubes and termite's as well as ant's nests. *Typhlops neszoelyi* (Daudin), a species of family Typhlopidae is the most common worm snake variously known as braminy blind snake, flowerpot snake, common blind snake etc. Since these three families are placed within the infraorder Scolecophidia of the order Ophidia, blind snakes are sometimes also called "Scolecophidian snakes". As such, the diversity quantum of such creatures although may not be threatened or endangered is very less. In this communication, the unique scattered patterns of distribution & low diversity in terms of species relative abundance on the basis of survey & sampling of the creature in their natural abode have been discussed.

Keywords:- worm snake, typhlopoidea, distribution & relative abundance

INTRODUCTION

Superfamily Typhlopoidea belonging to the infraorder Scolecophidia, suborder serpentes or ophidia of class reptilia is represented by several worms like nonpoisonous snakes that fall under three important taxonomic families viz Anomalepidae, Leptotyphlopidae and Typhlopidae.¹

There are several species of snakes which looks like worms and hence known as worm snake. These worms snake usually shares the characteristics of small size,

primarily subterranean habitat, and small eyes and thus resembles like earthworm and therefore sometimes people get confused. These snakes are sometimes also known as blind snake which is also alternative common name. Since these three families fall within infraorder Scolecophidia, blind snakes are sometimes also called "scolecophidian snakes". Blind snakes are mainly small with blunt heads, cylindrical bodies, and short tails. Their bodies are encased in smooth, shiny scales. All are fossorial and are regularly associated with termite and ant nests having scattered but clumped distribution. As such, many numbers of individuals of a species can be noticed in small clumped coil.²⁻³

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In the present communication the pattern of distribution of different species of the genus *Typhlina* belonging to the different family Typhlopoidea along with its low population density & abundance has been reported which may serve as the indicator of their population density in terms of survival security as well as management in their contribution in the environmental ecology.⁴

As regards the study distribution and abundance of these snakes, it is mandatory to understand their natural place of occurrence & feeding habits. Usually, they are found in both urban and rural agricultural areas with sufficient moisture in the soil.⁵ These species are terrestrial, but some specimens have been observed in arboreal habitats.⁶ These snakes also live underground in group (clumped form) in ant and termite nests which are called antorium and termitorium. They are also found under logs, moist leaves, stones and humus in wet forest, dry jungle, abandoned buildings and even city gardens. The distribution and survival of this group of blind snakes directly reflect soil humidity and temperature.⁷ Their diet consists of the larvae, eggs, and pupae of ants and termites.⁵

MATERIAL & METHOD

Random sampling of the snakes were made from different subterranean tubes & soil surface of following five places of Madhepura-Gamharia, Singheswar, Gwalpara, Kumarkhand & Alamnagar. In most of the habitat very low abundance of the species has been found, somewhere as low as one individual in one quadrat.

Table 1: Site specific abundance of three sampled species of worm snake from different sampling sites of Madhepura

Sl. no.	Name of the species	Individual no. per quadrat	Sampling sites
1	<i>Typhlops bramini</i>	4	Gamharia, Singheswar
2	<i>Typhlops loveridgei</i>	2	Gwalpara, Kumarkhand,
3	<i>Typhlops neszoelyi</i>	1	Alamnagar

Fig 1(A-E): Distribution & abundance of worm snakes from different sampling sites of Madhepura, Bihar



A. Species 1 in site 1: Gamharia (*Typhlops bramini*)



B. Species 1 (repeat) in Site 2: Singheswar (*Typhlops bramini*)



C. Species 1 in Site 3: Gwalpara (*Typhlops loveridgei*)



D. Species 2 (repeated) in Site 4: Kumarkhand (*Typhlops loveridgei*)

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E. Species 3 in Site 5: Alamnagar (*Typhlops neszoelyi*)

RESULTS & DISCUSSION

As per random sampling data obtained from this study, three different species of worm snakes belonging to the genus *Typhlops* have been found with single and repeated occurrence in five different sampling sites of Madhepura.(Table:1).

The number of individuals from each sampling quadrat reflects that *Typhlops bramini* was in highest abundance (4 in five sites) whereas *Typhlops neszoelyi* was in lowest (only 1 in site 5).

No such abundance report of the worm snake has been made by any author so far although world wide occurrence of various species of different genera like *Typhlops*, *Grypotyphlops*, *Ramphotyphlops* has been reported by various authors.

As regards distribution of the worm in the light of its natural occurrence in the habitat, it has been clearly observed that the snakes love single living in the open habitat while in clumped cluster in the ant & termite closed nests.

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Physico-chemical analysis of Kosi river water at Tengraha village of Supaul district

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Abstract : Water samples from some chours (water reservoirs) was collected and physico-chemical analysis was performed. pH value ranged in between 7-20 to 10, BOD 2mg/l to 40mg/l, COD between 11mg/l to 16mg/l. Water quality of these chours was moderate.

Keywords: Chours, BOD, COD

INTRODUCTION

Water is one of the most important compounds of the ecosystem. All living organisms need water. But due to industrialization, increased human population, urbanization, use of fertilizers and pesticides, human activities water reservoirs and river water is being polluted. Presence of hazardous material in water affects all animals, plants and human beings. There for it is necessary to check the water quality at regular intervals.

The availability of good quality of water is an indispensable feature for preventing diseases and improving quality of life. Natural water contains different types of impurities. Dissolution of aerosol particles from the atmosphere and from several human activities such as mining, addition of industrial slag, washing of clothes, bathing of animals, addition of municipal waste in water courses water pollution.

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Therefore, it is essential to test the water before it is used for drinking, domestic, agriculture or industrial purpose. Selection of parameter for testing of water depends mainly upon for what purpose water is used.

Some physical parameters such as pH, turbidity, temp. and chemical parameters such as BOD, COD, DO, chloride etc.

MATERIAL AND METHOD

A survey was conducted at Tengraha village of Supaul district from March 2012 to February 2013 at regular intervals.

Water samples were collected in 1 litre plastic bottles and preserved for laboratory analysis as per standard procedures recommended in APHA (1975). The preserved samples were brought to the laboratory and were kept at 4°C in refrigerator before the start of chemical analysis. The time interval between collection, storage and analysis was intentionally shortened to avoid any major change in chemical characteristics of the sample.

The analysis was done for temp., pH, Transparency, Carbonate, Bicarbonate, DO, BOD, COD, Chloride, Silica, Nitrate, Ammonia, Sulphide, Sulphate and Phosphate.

RESULT

The result is mentioned in Table below

Range of physico-chemical parameters at Site I (Tengraha) during 2012-13.

Parameter	Mar.12	Jun.12	Jul.12	Oct.12	Nov.12	Feb.13
	Min.	Max.	Min.	Max.	Min.	Max.
Temp.	28.00	32.00	18.00	18.00	28.00	30.00
pH	8.20	7.20	10.00	10.00	9.20	8.30
Transparency	16.00	15.00	80.00	80.00	11.00	21.00
Total Solids	110.00	109.00	68.00	68.00	104.00	107.00
Carbonate	14.00	16.00	13.00	13.00	13.00	18.00
Bicarbonate	225.00	170.00	165.00	165.00	179.00	155.00
DO	09.00	10.00	10.00	11.00	9.00	10.50
BOD	5.00	6.00	2.00	2.00	6.00	40.00
Chloride	70.00	80.00	74.00	74.00	75.00	80.00
Silica	8.00	8.50	6.00	6.00	6.50	6.00
Nitrate	0.70	0.80	0.95	0.95	0.75	0.85
Ammonia	1.60	0.80	1.30	1.30	0.870	0.90
Sulphide	0.40	0.80	0.60	0.60	0.80	0.70
Sulphate	2.00	2.20	0.75	0.75	1.20	0.85
COD	16.00	14.00	11.00	11.00	13.00	11.00
Phosphate	0.30	0.10	0.15	0.15	0.10	0.15

CONCLUSION

The water quality of these site was moderate - pH Ranged from 7.20 to 10, BOD and COD ranged from 2mg/l to 40mg/l and 11mg/l to 16mg/l respectively.

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Vitamins A, D and E: prospects in SARS- CoV2 therapeutics

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Abstract : The SARs-Cov2 pandemic commonly known as Covid-19 has affected the world like never before in the recent past. At present a highly efficient drug is yet to be found which can control and completely stop further damage to human health and world economy. A need for stopping further spread of this pandemic in minimal time using drug repurposing without long term experiments has surfaced. This study was thus designed to find possible targets for covid-19 therapeutics. We did docking (Insilco) study to find the usefulness of three commonly used vitamins for this purpose. Spike receptor binding domain without receptor was chosen as the target and the Vitamins A, D and E were the ligands. We found that the vitamins A and D were good candidates for covid19 therapeutics.

Key words: SARs-Cov2 pandemic, Covid-19, Vitamins A, D and E

INTRODUCTION

In the past century 1918 influenza and human immunodeficiency virus (HIV) pandemics have already baffled the world. In the present century the severe acute respiratory syndrome (SARS-CoV) and Middle Eastern respiratory syndrome corona virus (MERS-CoV) had already challenged the health system. The Ebola outbreak and severe acute respiratory syndrome corona virus-2 (SARS-CoV2) in the recent past have also raised the need for better public health management systems and drugs that could effectively treat the infected and vaccines for prophylaxis. The discovery of drugs is a boon to mankind. In present times the use of software's for designing new drugs and repurposing already available (which happen to be more than two billion) has made it possible to be done

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within time frames. The novel as well existing compounds are pitted against specific target proteins. This is done by evaluation of the protein-ligand affinity thermodynamics using different model systems.¹ This helps in avoiding far end research with inappropriate compounds as the chances of desired results with such compounds is very low. However, a particular compound which does not yield good result against one protein may be a better drug for another target protein. The free energy, enthalpy and binding energies are taken into consideration for selection of the best fit compound for a target protein. Further the ADMET properties of a compound help in screening only those molecules which will have tolerable adverse effects upon administration. This helps in decreasing the cost of drug designing also helps in speeding the transfer of a drug from bench to target population.²⁻⁴ The need and usefulness of such methods has been evident many times. The Ebola virus outbreak could be curbed because of success achieved in

treating patients with ebola virus by administrating Remdesvir. Such a success could not be achieved for SARS-CoV2 so far. Only a few drugs including Remdesvir, hydroxychloroquine.⁵⁻⁶ The importance of Vitamin D in limiting the fatality in SARS-CoV2 infected patients has been remarkable. It has already been shown that the patients of Caucasian race showed better survival than those the African origin. Studies are already in progression to find the efficacy of Vit. D administration for prophylaxis and treatment.⁷⁻¹³ The present study is thus an attempt to find the binding affinity of a three commonly used vitamins i.e., vitamins A, D and E to the spike receptor binding domain (SRBD) of SARs-Cov2.

The effect of Vitamin D and A has been highlighted by previous study.¹⁴⁻¹⁶ It acts on the expression of many genes through Vitamin D receptor. The complex of active vitamin D, VDRE retinoid x receptor and 9 cis retinoic acid interacts with vitamin D response elements and Vit D receptor genes. Further vitamins D and A interacts and modulates the functions of respiratory tract mucous membrane a site that is exposed to pathogens. Vitamin D controls the immune system by controlling the functions of WBCs and associated components. Of further importance is cathelicidin, a product of vitamin D stimulation which is antimicrobial in function.¹⁷⁻¹⁹

METHODOLOGY

The spike protein receptor binding domain without ace2 (Figure 1 (a) PDB id 6moj) was used as a target protein whereas the vitamins A and D were the ligands. The protein structures were downloaded from Protein Data Bank (PDB) and ligands were downloaded from pubchem database (Table 1) in PDB format. The files were uploaded to the web server DOCKTHOR. The protein and ligand were prepared and docked followed by analysis of the generated model systems. The results were downloaded.

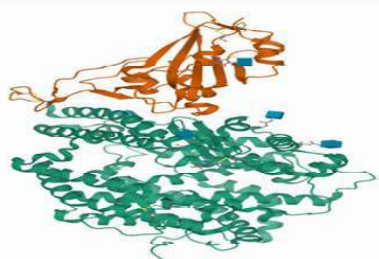


Figure a. The crystal structure of SARS-COV2-Receptor binding domain of spike protein (PDB code -6MOJ)

RESULTS AND DISCUSSION

There are no highly efficient drugs or vaccines available as of now for COVID-19 treatment and prophylaxis. We have to rely on repurposing of existing antiviral drugs available for treatment of several viral diseases with combination of vitamins (Vit.). In the present theoretical study we tried to find out the best fit ligand by doing insilico analysis. Vitamins, A E and D showed docking scores of -8.274, -8.350 and -7.452 respectively (Table 2). The total energy of the It has been shown in previous work that Vit. E did not show any significant effect in reducing respiratory tract infections, nor it shows any promising effects on reducing the severity of such infections. Instead of reducing it increase morbidity related to respiratory tract infections (RTI) further, Vit. E may compromise the efficacy of Vit. D and Vit. A based immune responses. In our present study we have found the binding energy and total energy of vit E with spike protein is higher (Table) than the binding energy of Vit. D and Vit. A. Hence more studies required to evaluate further the response of Vit E administration on covid patients either in combination with other vitamins or with drugs available for RTI. Our findings are similar to the findings of Masoudi *et al.* (2020)¹⁹ using MOE 2105 and Autodock vina for the affinity of vitamin D to 3CLpro an RNA dependent RNA polymerase.

CONCLUSION

We have found three vitamins which showed H- bond interactions with spiked without ACE2 receptor. The Vitamins A and D which also form complex that modulates the immune system are of importance. The administration of vitamin D in combination with vitamin A is an important aspect which needs to be explored further for usefulness in prophylaxis and improving the chances of recovery in Covid 19 patients.

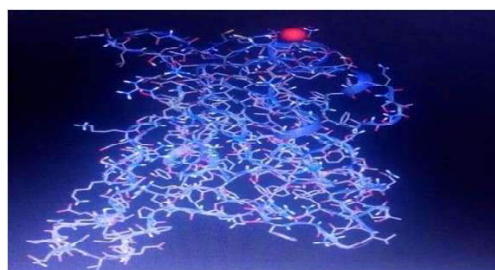


Figure b. SARS-COV2-Receptorbinding domain of spike protein after preparation on Dockthor webserver.

Table 1. Shows the Ligands and their 2D structures

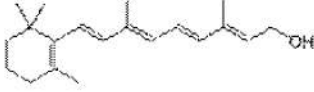
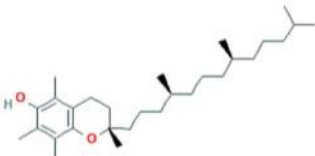
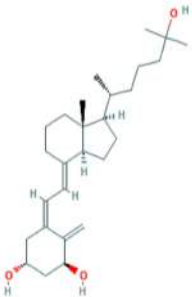
Sl. No.	Chemical structure			
1	Name	Retinol	Tocopherol	Calcitriol
2	Souce/Id	PubChem CID: 445354	PubChem CID: 14985	PubChemCID: 5280453

Table 2. Shows the binding affinity and total energy for the docking to the Spike receptor binding domain without ACE2.

Sl. No.	Ligands	Score	Total Energy	Vander Waal's	DockThor Job-id
1	Retinol	-8.274	98.72	-15.634	spikeminusaceretin_5f95238334442
2	Tocopherol	-8.350	303.323	-14.365	spikeminusacetcho_5f9511dcd6602
3	Calcitriol	-7.452	93.619	-14.621	spikeminusacecalcitriol_5f9504194ee3c

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Ethnomedicinal plants used for immunity booster to fight against SARS CoV-2 virus of COVID-19 by ethnic people of Gumla district of Jharkhand

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Abstract : India is one of the countries to have recorded the first case of COVID-19 in January 2020 and hence, is among the nations, most vulnerable to coronavirus epidemic. Despite huge and multiple challenges, India has been demonstrating unwavering commitment in its fight against the pandemic these actions have been lauded by WHO.¹ SARS CoV-2 stands for “Severe acute respiratory syndrome coronavirus 2” is the strain of coronavirus that causes respiratory illness responsible for the COVID-19 pandemic.^{2,3} E gene of the virus is detected from the sample of covid-19 patient. The coronavirus E protein is a small membrane protein that has an important role in the assembly of Virions where it may induce membrane curvature or aid in membrane scission.⁴ This article deals with the several plant species used for immunity booster by ethnic people of Gumla district and 20 plants and their way of administration has been described in the present paper.

Key words: COVID-19, SARS CoV-2, Pandemic, E gene

INTRODUCTION

Jharkhand is a state in eastern India. The state shares its border with the states of Bihar to the north, Uttar Pradesh to the northwest, Chhattisgarh to the west, Odisha to the south and West Bengal to the east.⁵ Gumla is in the southern part of the Chota Nagpur Plateau, which forms the eastern edge of the Deccan Plateau. It is predominantly a tribal district.⁶ The first case of the COVID-19 pandemic was confirmed in the Jharkhand on 31st March as a Malaysian woman came positive on the test.⁷ Now a day this virus is spreading very quickly and more and more cases confirms

daily. Symptoms may or may not be detected among infected persons. E gene of the virus is detected among people of infected ones.⁴ Till date there is no vaccine or treatment available for this pandemic so many plants are using for boosting immunity for fight against it. Enhancing the body's natural defense system (immunity) plays an important role in maintaining optimum health.⁸ Older adults are at risk for malnutrition, which may contribute to their increased risk of infection.⁹ Food rich in proteins, glucose, vitamin C and D are recommended for faster recovery from viral infections.¹⁰⁻¹² The present paper contributes to Ethnomedicinal Botany regarding plants having improving immunity power of the body.

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MATERIALS & METHODS

Various kinds of allopathic drugs have been used now a day for boosting immunity system of our body but they result in kinds of side effects thus the traditional method of using plants for immunity booster is highly appreciated by people of Jharkhand state. Different plant species were collected from different blocks and villages of Gumla

district via conversation with tribes. Firstly, field trips were conducted and local Vaidya's were contacted. All information regarding such plants was gathered. Detailed information regarding Botanical identification plants, local name, parts used in the medicine was recorded, herbaria was deposited with all gathered information in the DSPMU Ranchi.

Table 1: List of plants used for immunity booster and serve to fight against SARS CoV-2 of COVID-19 by ethnic people of Gumla district and their way of administration

Sl. No.	BOTANICAL NAME	FAMILY	COMMON NAME	PLANT PART USED
1.	<i>Oscimum basilicum</i>	Lamiaceae	Basil/tulsi	Leaves extract
2.	<i>Alium sativum</i>	Amaryllidaceae	Garlic/lahsun	Cloves. Active principal Allicin
3.	<i>Curcuma longa</i>	Zingiberaceae	Turmeric/haldi	Roots
4.	<i>Elettaria cardamomum</i>	Zingiberaceae	Cardamom/ilaychi	Seed
5.	<i>Trachyspermum ammi</i>	Apiaceae	Ajwain	Leaves and seeds
6.	<i>Citrus limon</i>	Rutaceae	Lemon/nimbu	Fruit
7.	<i>Zingiber officinale</i>	Zingiberaceae	Ginger/adrak	Tubers
8.	<i>Cinnamomum verum</i>	Lauraceae	Cinnamon/dalchini	Bark
9.	<i>Citrus reticulata</i>	Rutaceae	Orange/santara	Fruit
10.	<i>Cuminum cyminum</i>	Apiaceae	Cumin/jeera	Seeds
11.	<i>Withania somnifera</i>	Solanaceae	Ashwagandha	Roots
12.	<i>Trigonella foenum-graecum</i>	Fabaceae	Fenugreek/ methi	Seed
13.	<i>Tinospora cordifolia</i>	Menispermaceae	Giloy	Root/stem
14.	<i>Cinnamomum tamala</i>	Lauraceae	Bay leaf/Tejpata	Extract of leaf
15.	<i>Piper nigrum</i>	Piperaceae	Black pepper/kaalee mirch	Seeds
16.	<i>Phoenix dactylifera</i>	Arecaceae	Date palm/Jaggery	Fruit
17.	<i>Carica papaya</i>	Caricaceae	Papaya/papita	Fruit
18.	<i>Avena sativa</i>	Poaceae	Oats/jai	Seeds
19.	<i>Cucurbita pepo</i>	Cucurbitaceae	Pumpkin/kohda	Seeds
20.	<i>Helianthus annuus</i>	Asteraceae	Sunflower/surjmukhi	Seeds

RESULTS & DISCUSSION

The survey of many blocks and villages of Gumla district of Jharkhand revealed that the tribes are using many plants for boosting their immunity and for good health. After the survey we found that most of the herbal medicine are obtained from the nearby forest and are local wild weeds. They are taken mostly in the form of juice, pest, dry powder etc.

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Shannon-Weiner species diversity index of sunflower insect pest, *B.tabaci* (Hemiptera: Aleyrodidae) & allied species

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Abstract : This communication is furtherance of my previous work on the quantification of relative abundance of various insect pest species including *B. tabaci* on sunflower plant. The article deals with advanced statistical application to the prevailing population data of five different species of insect pests like- whitefly: *Bemisia tabaci* (Gennadius) (Hemiptera: Aleyrodidae) leafhopper: *Empoasca* spp. (Lepidoptera: Noctuidae), flower moth: *Thysanoplusia orichalce* (Lepidoptera: Noctuidae), bollworm: *Helicoverpa armigera* (Lepidoptera: Noctuidae) and stink bug: *Nezara viridula* (Heteroptera: Pentatomidae) reflecting composite pest infestation. These insects are taxonomically represented by the Hemipterans, Lepidopteran and Heteropterans. Advanced statistical tools like Shannon Weiner (S-W) species diversity index (**H**), relative abundance (RA%) and Jaccard's evenness(E) have been applied to the raw population data of pest species procured during the sampling in 2018. The results are very significant in the terms of pest species secured and protected survival on the host plant- sunflower at the latter's cost that blows serious alarm to urgently envisage the conservative strategies for the host plant being the revenue generating crop in Kosi basin agricultural land covering Saharsa Madhepura, Purnea etc of North Bihar.

Key words: Advanced statistical tools, S.W diversity index (**H**), RA %, Jaccard's Evenness(E), insect pest species, composite infestation, sunflower plant

INTRODUCTION

The whitefly *Bemisia tabaci* (Gennadius) is one of the most threatening pests to sunflower crop production in tropical and subtropical regions.¹ In this research communication, the results of advanced statistical diversity tools applied to the natural sampled population of different pest species have been presented to suggest envisaging the strategies for minimising the composite pest infestation and protecting the crop.²

S.W diversity index (**H**), RA %, Jaccard's Evenness (E) are some advanced statistical tools being used for

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interpreting the inferential impact of any species population whether beneficial or harmful on the environmental resources of both abiotic and biotic nature.³ All these considerations fall under portfolio function of ISI (Information Statistical Indices) helpful in holistic management and conservation of any community, habitat or biosphere reserve.

Being one of the major competitor to human beings insects may be both harmful as well as beneficial, therefore, insects are our rivals, because most of them lead pest life feeding on plants and crops.⁴ As such they are major threat to crop production and human survival.

In the present article advanced statistical information tools in the form of S-W diversity, Jaccard's Evenness (E)

& Simpson dominance equations (indices) have been used to compute the raw sampling data of five different insect pests compositely attacking the single host - sunflower, *Helianthus annuus* (L.)⁵ popularly known as “Surajmukhi” belonging to the family Asteraceae. It is one of the most important oil crops throughout the world grown on over 22 million hectares worldwide, with a production of 26 million tons per year.^{6,7}

MATERIAL & METHODS

Selection of the sampling site as follows up schedule of our study was done randomly and the sites were located in the northern zone of Madhepura district. Madhepura is located at 25°34’-26°07’ North Latitude & 86°19’-87°07’ East Longitude. It is surrounded by Araria & Supaul in the North and Khagaria & Bhagalpur in the South, Purnea district in the East and Saharsa in the West. It is situated in the plains of river Koshi & located in the Northeastern part of Bihar. The primary occupation of the people of the district is agriculture.⁸ The total geographical area of the district is 1788 Sq. Km.

Random sampling of different pest flies infesting the sunflower plant by net-sweeping method:

The field experiment was laid in randomized block design during 2018 in four different areas of Madhepura where the production was very high. The observation data of random sampling of different insect pest flies infesting the host with respect to the four different sampling areas - Gwalpara, Puraini, Bihariganj and Shankarpur has been given in Table 1. Additionally, observations of infestation on randomly selected plants from each plot by whitefly and other allied species were made on leaves shoot and flower rosette. The mean population of each pest species

per plant in a quadrat plot was counted numerically & furnished in the data recording table for advanced statistical analysis.

RESULT & DISCUSSION

The computation results of five pest species population infesting sunflower plants as found in the random sampling data in four different plots with regards to S-W species diversity, relative abundance, species richness being the log value of the abundance proportion & Jaccard's Evenness are very informative which throw light on the sustainable relationship between the host & pest. As such all the pest species are nutritionally well supported by the host sunflower plants. However, the strategic efforts to be taken up for the pest control & protect the host require to break this statistical bondage between the host & the pest which reinforced by the demand & supply of the nutritional molecules & the elements of pest & host respectively culminating into the sustainable relationship. The results of statistical analysis of site specific sampling data with respect to R.A, S-W diversity & Evenness reflect conspicuous variations which are denoting the similar variation in the density dependent availability of nutritional resources from the host. These statistical variation also indicate the gradient in the environmental resources other than biotic factors like climatic constitution of the site including temperature, humidity soil condition etc.

The collective discussion on the intricate relationship between the results of statistical species diversity indices & environmental as well as the host resources demands special consideration in the terms of asset management terms of the nature falling under portfolio management of the living community.⁹

Table 1: Site specific advanced statistical information table regarding the cumulative species diversity of five different insect pests infesting sunflower plant (Sampling site A-Gwalpara)

Sl. no	Common name of the insect pest sampled	Scientific name as per the key	Taxonomic position	Number of adults per five plants (n)	Relative abundance % (n/N* 100)	Individual mean abundance (pi)	Log value of (log pi)	Product of pi*log pi	S-W diversity (H)	Jaccard's evenness (E) = H/Hmax
1	Whitefly	<i>Bemisia tabaci</i>	Hemiptera : Aleyrodidae	12	36.3	0.363	-0.440	-0.1597	0.6317	0.935
2	Leaf hopper	<i>Empoasca</i>	Lepidoptera Noctuidae	04	12.1	0.121	-0.917	-0.1109		
3	Flower Moth	<i>Thysanoplusia</i>	Lepidoptera Noctuidae	02	6	0.060	-1.221	-0.0732		
4	Bollworm	<i>Helicoverpa</i>	Lepidoptera Noctuidae	06	18.1	0.181	-0.742	-0.1343		
5	Stink bug	<i>Nezara</i>	Heteroptera: Pentatomidae)	09	27.2	0.272	-0.565	-0.1536		
	Total			N =33						

Kumari- Shannon-Weiner species diversity index of sunflower insect pest, *B.tabaci* (Hemiptera: Aleyrodidae) & allied species

Table 2: Site specific advanced statistical information table regarding the cumulative species diversity of five different insect pests infesting sunflower plant (Sampling site B- Puraini)

Sl. no	Common name of the insect pest sampled	Scientific name as per the key	Taxonomic position	Number of adults per five plants (n)	Relative abundance % (n/N* 100)	Individual mean abundance (pi)	Log value of (log pi)	Product of pi*log pi	S-W diversity (H)	Jaccard's evenness (E) = H/Hmax
1	Whitefly	<i>Bemisia tabaci</i>	Hemiptera : Aleyrodidae	10	31.2	0.312	-0.505	-0.157	0.65	0.962
2	Leaf hopper	<i>Empoasca</i>	Lepidoptera Noctuidae	05	15.6	0.156	-0.806	-0.125		
3	Flower Moth	<i>Thysanoplusia</i>	Lepidoptera Noctuidae	08	25	0.25	-0.602	-0.150		
4	Bollworm	<i>Helicoverpa</i>	Lepidoptera Noctuidae	02	6.2	0.062	-1.207	-0.074		
5	Stink bug	<i>Nezara</i>	Heteroptera: Pentatomidae)	07	21.8	0.218	-0.661	-0.144		
	Total			N=32						

Table 3: Site specific advanced statistical information table regarding the cumulative species diversity of five different insect pests infesting sunflower plant (Sampling site C- Bihariganj)

Sl. no	Common name of the insect pest sampled	Scientific name as per the key	Taxonomic position	Number of adults per five plants (n)	Relative abundance % (n/N* 100)	Individual mean abundance (pi)	Log value of (log pi)	Product of pi*log pi	S-W diversity (H)	Jaccard's evenness (E) = H/Hmax
1	Whitefly	<i>Bemisia tabaci</i>	Hemiptera : Aleyrodidae	9	31	0.310	-0.508	-0.1574	0.6626	0.981
2	Leaf hopper	<i>Empoasca</i>	Lepidoptera Noctuidae	03	10.3	0.103	-0.987	-0.1016		
3	Flower Moth	<i>Thysanoplusia</i>	Lepidoptera Noctuidae	08	27.5	0.275	-0.560	-0.154		
4	Bollworm	<i>Helicoverpa</i>	Lepidoptera Noctuidae	05	17.2	0.172	-0.764	-0.1314		
5	Stink bug	<i>Nezara</i>	Heteroptera: Pentatomidae)	04	13.7	0.137	-0.863	-0.1182		
	Total			N=29						

Table 4: Site specific advanced statistical information table regarding the cumulative species diversity of five different insect pests infesting sunflower plant (Sampling site D- Shankarpur)

Sl. no	Common name of the insect pest sampled	Scientific name as per the key	Taxonomic position	Number of adults per five plants (n)	Relative abundance % (n/N* 100)	Individual mean abundance (pi)	Log value of (log pi)	Product of pi*log pi	S-W diversity (H)	Jaccard's evenness (E) = H/Hmax
1	Whitefly	<i>Bemisia tabaci</i>	Hemiptera : Aleyrodidae	11	30.5	0.305	-0.515	-0.1570	0.6751	1
2	Leaf hopper	<i>Empoasca</i>	Lepidoptera Noctuidae	08	22.2	0.222	-0.653	-0.1449		
3	Flower Moth	<i>Thysanoplusia</i>	Lepidoptera Noctuidae	07	19.4	0.194	-0.712	-0.1381		
4	Bollworm	<i>Helicoverpa</i>	Lepidoptera Noctuidae	04	11.1	0.111	-0.954	-0.1058		
5	Stink bug	<i>Nezara</i>	Heteroptera: Pentatomidae)	06	16.6	0.166	-0.779	-0.1293		
	Total			N=36						

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Prolonged colchicine induced nephrotoxicity in albino rat

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Abstract- As follow up of previous investigation on effect of prolonged use of colchicine alkaloid on the hepatic tissues (hepatotoxicity) in albino rats, the present communication deals with the picture of nephrotoxicity in terms of renal function tests result carried out in the same subject simultaneously experiencing the prolonged treatment of intraperitoneal injection of different concentration of colchicine for a period of three months. The observed biochemical assay of centrifused blood serum of treated rats for renal function test were serum creatinine (SC) & serum urea (SU) as per the standard protocol of Wilding & Kennedy (1977)¹ and Alexander Griffith (1992)². Results obtained have been found to be statistically significant, establishing the toxic impact of the alkaloid colchicine on the kidney of albino rats.

Key words: Nephrototoxicity, renal function tests, biochemical assay, SC & SU, Colchicine, prolonged use

INTRODUCTION

Alkaloids are group of naturally occurring compounds containing nitrogen, oxygen, carbon & hydrogen primarily found in flowering plants. Montanine, morphine, colchicine etc. are among the major groups of alkaloids having various specialized pharmacological activities. Colchicines are considered as one of the weak anti-inflammatory activity derived from the autumn crocus *Colchicum autumnale* and the glory lily *Gloriosa superba*. It also functions as anti-mitotic agent & is helpful in preventing various types of diseases such as gouty arthritis, Mediterranean fever & also causes depolymerization of microtubules.³

Overdose of colchicine constitutes a toxicological emergency and rapid intervention. Generally colchicine

also inhibits the multiple proinflammatory mechanism while increases the level of anti-inflammatory mediators. There are various suggestions to explain the effect of colchicine. It is thought that there may be a direct toxic effect on the myocardial cells with impairment of impulse generation and cardiac conduction.^{4,5} This has not been proved, although a similar mechanism of direct toxicity is seen on the cells of skeletal muscle.⁶ It is also possible that the profound acid base disturbances and electrolyte derangements associated with overdose plays a significant part.⁷ Due to the vital importance of colchicine this work was undertaken to investigate whether colchicine administration to normal rats has any histological effects on the kidney as well as the other tissues. Colchicine has been reported to protect against a variety of hepatotoxic insults. It was reported to improve survival in a clinical trial for alcoholic cirrhosis.⁸ Thus the paper deals with the study of nephrotoxicity of colchicine in rat.

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MATERIALS & METHODS

For the analysis of nephrotoxicity of colchicine thirty albino rat weighing about (200–250 g) were considered for the study. All the thirty rats were kept in a laboratory under standard condition and were fed with normal food and water. The animals were then divided into three groups each having three individuals each. Group 1 of the mice were named as control group that was only fed with normal distilled water & food, group 2 of the animals were given 2mg of colchicine/kg/day while group 3 of the animals were fed with 3 mg/kg/day of colchicine dissolved in the distilled water by intraperitoneal injection.

After 24 hours the animals were then sacrificed and sample of the blood was taken from each of the animals and centrifuged at 3000 rpm to separate the serum. The separated serum was then subjected to biochemical analysis for following kidney function tests:

Renal function tests:

1. Serum creatinine

It was determined by Jaffe reaction.¹

2. Serum urea

It was determined by Berthelot method.²

Ethical consideration

For the experiment most appropriate animal was chosen for the research. Great care was taken at each times of the experiment. For obtaining statistically the valid result animal of appropriate size & fewer numbers were for the used. Painful procedures were performed under anesthesia to avoid distress and pain. Our standards of animal care and administration met those required by applicable international laws and regulations.

RESULT & DISCUSSION

Renal function tests:-

The result obtained from the above experiment has been shown in the Table no 1 that shows mean +SD values of rats renal function tests (serum urea and creatinine). For the control group mean +SD values of urea were 12±1.1 which received only distilled water, group 2 animals which received 2 mg/kg/day of colchicine dissolved in distilled water the mean +SD values was 30.3±2.2 and 45.2 ± 2.4 in the third group which was received 3 mg/kg/day of colchicine dissolved in distilled water. For creatinine the mean +SD values of were

0.32±0.5 in the control group which received distilled water, 0.59±0.1 in the second group which received 2 mg/kg/day of colchicine dissolved in distilled water and 0.97±0.1 in the third group which was received 3 mg/kg/day of colchicine dissolved in distilled water.

Table 1: Effect of prolonged use of colchicine on Mean + SD of renal function tests in rats

Group	Group 1	Group 2	Group 3
Urea	12±1.1	30.3±2.2	45.2 ±2.4
Creatinine	0.32±0.5	0.59±0.1	0.97±0.1

DISCUSSION

Colchicine which is one of the plant alkaloid functions as a safe drug when taken in appropriate dose. But when taken in heavy amount it may lead to various life threatening diseases. So one of our most important function of our present experiment was to study about the toxic effect of colchicine in the kidney functions of the rat i.e. the nephrotoxicity of the colchicine.

The present study indicated that there is a statistical significant increase of serum urea and creatinine in third group which received 3mg/kg/day of colchicine in comparison to control group and second group which received 2 mg/g/kg/day of colchicine. It was also confirmed by other authors that renal failure develops a result of direct toxic effect of colchicine on the renal tubules.⁹

Our study confirmed that prolonged use of colchicine with high different doses leads to disturbance of renal physiological function which depends on pathological condition of the kidney. Increase of serum creatinine and urea due to functional capacity disturbance of tubular excretion, nuclear and mitochondrial protein function change.¹⁰ These changes results from oxidative stress generating reactive oxygen species induced by colchicine on the kidney depending on its dose. Thus prolonged use of colchicine leads to nephrotoxicity.

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Histopathological studies of some insecticides on the alimentary canal of Khapra beetle *Trogoderma granarium* (Evert)

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Abstract- The present investigation describes the histopathological changes of different sections of alimentary canal of *Trogoderma granarium* (Evert) due to effect of experimentally ascertained LD₅₀ doses of Gamma-BHC, Malathion and Fenitrothion. The uses of insecticides for effective and prompt control of the harmful population of insects are widely prevalent now a day. Their extensive use in pest management is mainly due to their simplicity, promptness, effectiveness and flexibility in character. The histopathological study of alimentary canal of different insects due to the effect of the insecticides has been studied by various workers such as Pilat (1935)¹, Woke (1940)², Eckert (1948)³, Lal *et al.* (1970)⁴, Misra and Mukherji (1972)⁵, Misra *et al.* (1972)⁶ & Mukherji and Misra (1972)⁷. In the present investigation, histopathological changes of alimentary canal of *Trogoderma granarium* (Evert) with Gamma-BHC, Malathion and Fenitrothion have been studied.

Key words: Khapra beetle, Pest, Alimentary canal, LD₅₀ doses, Insecticides, Maize.

INTRODUCTION

Insect is the oldest animal in this world, found almost everywhere, often in huge numbers. As the Khapra beetle is one of the most feared stored product pests. Population of this pest can build rapidly in the short time under hot and dry conditions. Grain damage depending on existing conditions often reaches upto 70% damage has been reported. It prefers grain and cereal product particularly maize (*Zea mays*). Generally insects may annoy for human beings but on the other hand, they are very helpful in various ways for instance in removing dead animals, producing honey, wax, lac, silk etc. It is roughly estimated that the annual loss due to various pests affecting our crops

is about one thousand five hundred crores rupees.⁸ However, insects are of great significance to mankind because they affect the human economy to a large extend. Therefore, the annual loss to the stored grains is estimated to be around upto 20% due to inadequacy of storage. The major insect pest of stored grain is *Trogoderma granarium* (Evert). Its tissues are selected for the investigation of the foregut, midgut and hindgut. The present investigation deals with the histopathological changes induced by the selected insecticides on the different parts of the alimentary canal of the maize pest *Trogoderma granarium* (Evert).

MATERIAL & METHODS

Laboratory culture of the insects was raised on maize (*Zea mays*) at 30±2°C temperature and 70 ±5 % R.H. only newly immersed adult insects were used in the experiment.

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The insecticides used in the investigation were the technical grades of Gamma-BHC, Malathion and Fenitrothion. Experimentally ascertained LD₅₀ doses of Gamma-BHC, Malathion and Fenitrothion were administered into the alimentary system of each insect by means of micro drop applicator. The moribund stage of the test insects reached after about 10 hours of the treatment. These insects were fixed in the alcoholic Bouin's fixative after cutting the legs and elytra with the help of a very sharp blade. Paraffin blocks were prepared followed the standard procedure of dehydration and paraffin embedding at 58°C. During final stage of dehydration and clearing, instead of absolute alcohol and xylene, 96% alcohol and amyl acetate were used. Sections were cut at 8µ. The usual double stain technique of haematoxylin and eosin was used for staining the sections. Control slides of untreated insects were also prepared by followed the same procedure.

OBSERVATIONS

The following histopathological changes were observed in the different sections of alimentary canal of adult *Trogoderma granium* when treated with LD₅₀ value of gamma-BHC, Malathion and Fenitrothion.

(1) Effects on the Foregut:

No histopathological changes were observed in the section of alimentary canal due to the effects of insecticides used in the experiment.

(2) Effects on the Midgut:

(a) Gamma-BHC Poisoning:

Little changes were observed in the epithelial layer of the midgut. The epithelial cells showed signs of shrinkages of cellular cytoplasm. Due to this, a number of small vacuoles appeared in the cells. The boundaries, of the cells were not distinct due to shrinkages. At these places the shrinkage of cytoplasm resulted in the detachment of basement membrane from the epithelial cells. Nuclei of the epithelial cells were also displaced, hence not distinct clearly. No significant changes were detected in the muscular layer (Fig. no. -3).

(b) Malathion Poisoning :

(i) Upper part of the midgut:

The epithelial layer was greatly affected due to Malathion poisoning. These cells showed their complete detachment from the basement membrane. At many points,

due to longitudinal splits epithelial layer was broken into fragment were found to be thrown into lumen. In the broken fragment of epithelial layer, the nuclei were also found to be displaced. Muscle layers also showed signs of shrinkages (Fig. No.-4).

(ii) Lower part of the midgut:

The histopathological changes due to Malathion poisoning was not found to be the same throughout the length of the midgut. In the lower part of the midgut, different changes were observed. At a number of places, the epithelial layer got greatly folded due to shrinkage. As a result of this, the lumen was found to be greatly reduced. Large number of small vacuoles also appeared in the epithelial layer. This resulted due to the displacement of the nuclei. The basement membrane and muscle layers were greatly fused with the epithelial layer due to the shrinkage. (Fig. No. -5).

(c) Fenitrothion Poisoning:

Fenitrothion poisoning showed marked histopathological changes throughout the length of the midgut. The epithelial layer was found to be completely detached from the basement membrane. Due to longitudinal splits, this layer was found to be broken at many points. Epithelial cells were also found extended. Due to this, nuclei were displaced and moved more towards the basement membrane. The end of the epithelial cells towards the lumen became spindle-shaped. No significant changes were observed in the muscle layers (Fig. No. -6) .

(3) Effects on the hindgut:

This section of alimentary canal was found to be affected by Gamma-BHC poisoning. The cuboidal epithelial cells got folded and greatly shrank, as a result of which lumen were reduced. No significant changes were seen in the muscle layers (Fig. no. -7).

RESULTS AND DISCUSSION

Foregut of the alimentary canal was not affected by any of the three insecticides. The midgut region showed marked histopathological changes due to effect of all the three insecticides gamma-BHC only. Effects were of varying degree of vacuolations, detachments, splitting and shrinkages in the epithelial layer of midgut whereas, only shrinkages folding and thereby reduction in the size of lumen were found in the hindgut.

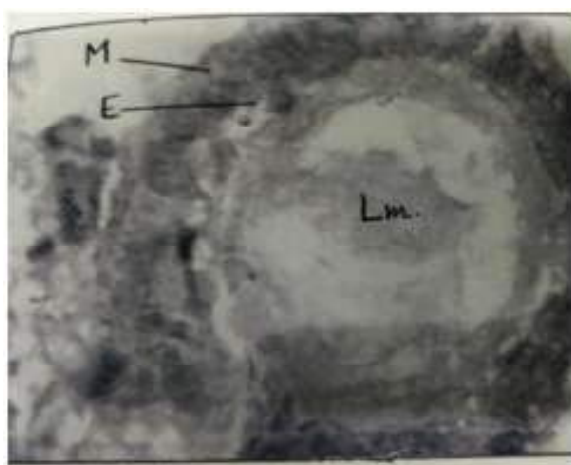


Fig . no. 1-T.S. of Midgut showing normal structure.

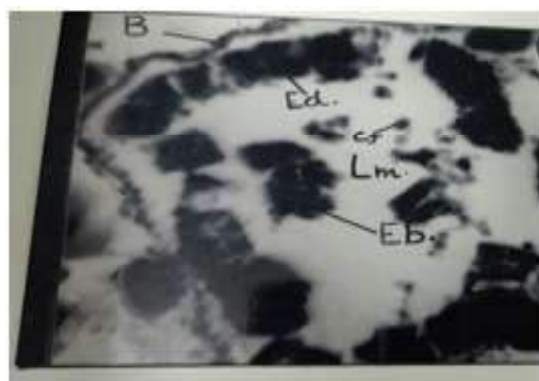


Fig. no. 4- .T.S. of upper part of the midgut treated with Malathion showing detached the broken epithelium (X 400).



Fig. no.2-T.S. of hindgut showing normal structure.



Fig. no. 5- T.S. of lower part of the midgut treated with Malathion showing greatly folded and vacuolated epithelium having displaced nuclei and reduced lumen (X 150).



Fig. no. 3- T.S. of midgut treated with gamma-BHC showing signs of shrinkage and detachment of epithelial layer (X 150).

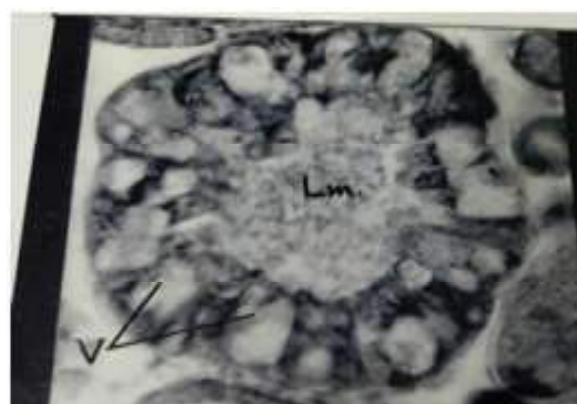


Fig. no. 6 – T.S. of midgut treated with Fenitrothion to show completely detached epithelium longitudinal splits and elongation of epithelial cells having displaced nuclei (X 400).

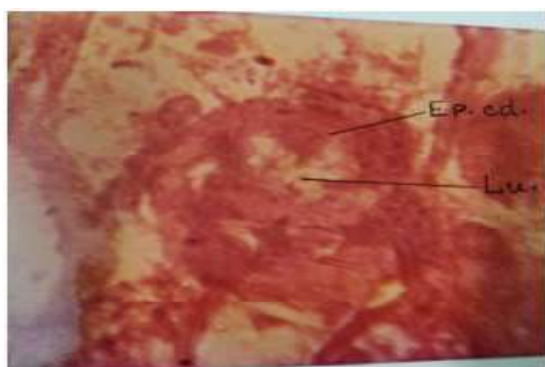


Fig. no. 7 – T.S. of hindgut treated with gamma-BHC showing greatly shrunk and folded epithelium and very reduced lumen (X150).

Similar findings on the foregut were found by Brown (1951)⁹ and Steinhans (1967)¹⁰. The cause of no marked histopathological change in this region may be due to the presence of intima and retention of the insecticides for a very little time in this region of alimentary canal. But on the other hand, marked histopathological change in this foregut, due to the effect of various insecticides have been observed by Sharma (1966,1968)^{11,12} in *Poecilercus pictus* and *Dysdercus koenigie*, Lal *et al.* (1970)⁴ in *Spodoptera litura*. Misra and Mukerji (1972)⁶ in *Drosophila melanogaster*.

Since the midgut epithelium is physiologically the most active site of the alimentary canal, most of the digestion of the food and absorption of the digested food take place in this region (Metcalf and March, 1953; O'Brien, 1957)^{13,14}. Muscle layers have shown the sign of shrinkages and their fusion with the basement membrane in the lower part of the midgut. Steinhans (1967)¹⁰ has suggested that the histopathological changes in the midgut are variable. Minor to serious damages of similar nature in the cells of midgut of *Vanessa uticae*, *Prothetria dispar* and *Pieris brassicae* have been also reported by Pilat (1935)¹. Woke (1940)² has observed disintegration of entire epithelial cells of the midgut in *Prodenia eridania* due to arsenic poisoning. Soliman and Soliman (1958)¹⁵ have also described the changes in the midgut in the larvae of *Prodenia litura* due to parathion poisoning. Mukherji and Hardas (1954)¹⁶ and Soliman and Shehata (1963)¹⁷ have studied the effect of parathion and other chlorinated hydrocarbons on second nymphal instar of *Schistocerca gregaria* and larvae of *Musca domestica* respectively. They

have reported the complete disintegration of midgut epithelium in both the cases. Hassanein and Khalil (1969)¹⁸ have reported elongation of the midgut of *Coccinella undecimpunctata* due to metasystox poisoning Topozada *et al.* (1968)¹⁹ have also reported the histopathological changes in the epithelial cells of the *Spodoptera littoralis*. Datta and Das (1971)²⁰ and Misra and Mukerji (1972)⁶ have reported similar findings in the midgut of *Periplaneta americana* and *Drosophila melanogaster* respectively. Sah and Srivastava (1989)²¹ have reported the histopathological changes of different sections of alimentary canal of pulse beetle due to the effect of experimentally ascertained LD50 doses of different insecticides. Although it is difficult to point out the exact cause for histopathological changes, but it can be concluded that with all probability, all the marked changes, such as, detachment of epithelial layers from the basement membrane, vacuolation and longitudinal splitting of the same, displacement of nuclei of the epithelial cells from their original position are due to shrinkage of the cells caused by the loss of water.

The changes in the hindgut have been observed only due to gamma-BHC poisoning. The changes may not be considered to be of much significance, when compared with that of the midgut. Because it causes only reduction in the lumen due to the folding and shrinkages of cuboidal epithelial cells. The vacuolation in the epithelial layer and its detachment from the basement membrane and the displacement of nuclei from their position are not seen. This may be due to the thickness of intimae in this region. The other two insecticides used in the experiment can be completely utilized in the midgut causing severe damage there, hence, they do not reach the hindgut in significant concentration to cause any damage to this part.

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Mortality of *Cirrhinus mrigala* exposed to different concentration of Sevin and Furadan

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Abstract- In the present study, during bio-assay tests, the percent mortality decreased and survival of the fish (*Cirrhinus mrigala*) decreased with an increase in the concentration of both the pesticides i.e. Sevin & Furadan (Carbofuran) as the fish exposed to lowest concentration of both pesticides showed a comparatively less reaction than that of the fish exposed to higher concentrations, while in highest concentration of both pesticides, rapid rate of mortality observed during the present study, might be due to their hyper-activeness and/or due to injury in some vital organs or due to some metabolic changes in the fish. The fish exposed to different concentrations of both pesticides showed abnormal behaviour depended on the concentration & exposure periods as hyper activeness & restlessness, vigorous swimming on the water surface, trying to jump out loss of equilibrium, uncoordinated movements followed by abnormal posture & swimming on one side, coma followed by mucus formation & death.

Key words: Mortality, *Cirrhinus mrigala*, Sevin, Furaden.

INTRODUCTION

For the last few years both organophosphate & Carbamate pesticides are used in place of organochloride pesticide and the carbamates are latest arrival in the field of cholinesterases pesticides. As present many new carbamate based pesticides are still in process of being marketed. These pesticides are least toxic than that of organophosphates and also bio-degradable in general, carbamate pesticide are synthetic derivative of physostigmine the principle alkaloids of the plant, *Physostigma venenosum* with rapid degradability nature. The most common and widely used carbamate pesticide

is carbaryl under the trade name of Sevin. It has low mammalian toxicity as an acute Le_{50} in rats is of 500-700 mg/kg body weight; whereas, the second, carbofuran with trade name of Fumdan ranks number two in U.S. list of carbamates to highly toxic with an acute oral Le_{50} value in rate of 5mg/kg body weight, These are powerful anticholinesterase agent. Operhuizen *et al.*, (1985)¹, Rao (1989)², Rao *et al.*, (2003)³, Vineet Kumar *et al.*, (2008)⁴ studied about lethality of different pesticides on fishes.

The aim of the present work is to study mortality/survival rate of in an important fresh water teleostean fishes exposed to Pesticide- Sevin and Furadan for different duration and in different concentrations in male and female fishes. Besides an attempt was also made to correlate the effect of each pesticide on changes with their mode of life.

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MATERIALS AND METHOD

Collected fishes were bathed for 15 minutes in 0.1 % aqueous potassium permanganate solution, followed by at least three wash in fresh ground water and then acclimatized for ten days in large aquaria/plastic tanks containing ground water in the laboratory. During this period, fish were provided artificial food prepared in the laboratory by following procedure at least three hours prior to change of ground water daily.

Stock solution of both Sevin & Furadan was prepared individually. A known amount of pesticides was first dissolved in minimum amount of Acetone (separately). The required concentrations were prepared by adding suitable aliquot of stock solution to ground water in a plastic tank/drum & finally transferred to test tanks. Acetone was used as a solvent since it is known to be non toxic up to 1,0 ml/1 concentration, whereas, the minimum amount of acetone in any concentration of the pesticides during this experiment was not more than 0.8 ml/l even in highest concentrations of the pesticides solution.

The total mortality at 24, 48, 72 & 96 hr. of exposure in each concentration of both pesticides were recorded. It is pertinent to mention here that the bioassay test of Sevin and Furadan was done on different days i.e. a gap between them was of ten days.

The percent mortality and probit kill of the fish in different concentrations of Sevin and Furadan were analysed by probit analysis method.

Probit analysis method: - In this method, percent mortality values are transformed in to probits with the help of "Probit table" and are plotted as ordinate.

RESULT AND DISCUSSION

It is known that when a pesticide or any pollutant is introduced into the environment there is a reasonable chance that it will ultimately finds its way in to the surrounding aqua-systems. Hence, aquatic system represents one of the most important complex environment as far as describing the fate & behaviour of the pollutants. Zooplanktons, comprising a large portion of living matter in natural waters, play an important role in biogeochemical cycles. Perhaps, these planktons can accumulate the toxicants at a higher level and at the same time, being primary consumers, they are assumed to be important

organisms transferring such accumulated toxicants to higher trophic levels especially to fishes.

It is well known from the works of Doudoroff & Katz (1953)⁵ & Khangarot & Ray (1987)⁶ that the physico-chemical properties of water have a definite role on the toxicity of heavy metals, but their effect on the toxicity of pesticides is relatively little known. Lloyd & Herbert (1962)⁷ have reported an increase in the toxicity of pesticide at higher temperature due to their metabolic rates. The resistance capacity of a fish to poison varies greatly between the individuals of the same species as larger fish have comparatively more tolerance power than the smaller fish of the same species (Weiss & Botts, 1957)⁸. Further, Larson *et al.* (1977)⁹ have stated that the resistance capacity of a fish to poison is depended on various factors like species, age, size, sex, life-cycle stages and metabolism.

Fishes were first explored by small scale exploratory bioassay test (APHA 1985)¹⁰. Then the logarithmic series of the doses were used to prepare various concentrations of the pesticides with the concentration factor of 1.25. Accordingly, ten fish (in each concentrations with control) were exposed to nine concentrations of Sevin ranged in between 10.0 mg/l to 59.6 mg/l and nine concentrations of Furadan ranging in between 6.0 mg/l to 35.7 mg/l. The whole experiments along with controls were repeated to see the changes if any.

As no mortality was observed in controls, the data were not subjected to Abbott's formula. The fish, which showed no respiratory movements & not responding prodding with a glass rod, were recorded dead and removed immediately as delay may affect tolerance limit of other alive fishes. Accordingly, the mortality of the fish in each concentrations were observed at three to four hours of interval upto 24 hrs & three or four times from 24 to 96 hr. The total mortality at 24, 48, 72 & 96 hr. of exposure in each concentrations of both pesticides were recovered and presented in Tables-1 and 2.

The toxicity of the chemicals is normally classified in to extremely toxic (1mg/kg body wt.) highly toxic (1 to 50 mg/kg body wt.), slightly toxic (0.5 to 5 gm/kg body wt.) and relatively harmless (more than 15 gm/kg body wt.). Further, several reports have appeared suggesting that, biocides have caused significant changes in metabolic process of fish (Vutukuru, 2005; Singh *et al.*, 2009)^{11,12}.

Yadav- Mortality of *Cirrhinus mrigala* exposed to different concentration of Sevin and Furadan

Table-1. Mortality of *Cirrhinus mrigala* exposed to different concentrations of Sevin at 24, 48, 72 and 96 hrs. exposure (expressed in both percentage and Probit kill after Finney, (1981)

Concentration (mg/l)	Log value of concentration	No. of fish exposed	Fish Mortality									
			24 hr.					48 hr.				
			No. of survival	No. killed	% survival	% kill	Probit kill	No. survival	No. killed	% survival	% kill	Probit kill
CONTROL	-	10	10	0	100	0	-	10	0	100	0	-
10	1	10	10	0	100	0	-	10	0	100	0	-
12.5	1.097	10	10	0	100	0	-	10	0	100	0	-
15.6	1.193	10	10	0	100	0	-	10	0	100	0	-
19.5	1.29	10	10	0	100	0	-	10	0	100	0	-
24.4	1.387	10	10	0	100	0	-	9	1	90	10	3.7184
30.5	1.484	10	9	1	90	10	3.7184	7	3	70	30	4.4756
38.1	1.581	10	6	4	60	40	4.7467	4	6	40	60	5.2533
47.7	1.678	10	3	7	30	70	5.5244	1	9	10	90	6.2816
59.6	1.775	10	0	10	0	100	8.719	0	10	0	100	8.719
Concentration factor = 1.25												
			72 hr.					96 hr.				
Control	-	10	10	0	100	0	-	10	0	100	0	-
10	1	10	10	0	100	0	-	10	0	100	0	-
12.5	1.097	10	10	0	100	0	-	9	1	90	10	3.7184
15.6	1.193	10	10	0	100	0	-	7	3	70	30	4.4756
19.5	1.29	10	9	1	90	10	3.7184	4	6	40	60	5.2533
24.4	1.387	10	6	4	60	40	4.7467	1	9	10	90	6.2816
30.5	1.484	10	2	8	20	80	5.8416	0	10	0	100	8.719
38.1	1.581	10	0	10	0	100	8.719	-	-	-	-	-
47.7	1.678	10	-	-	-	-	-	-	-	-	-	-
59.6	1.775	10	-	-	-	-	-	-	-	-	-	-
Concentration factor = 1.25												

Average weight of the Fish: 50±3 gm

Temp. 25.3±2.7°

Table-2. Mortality of *Cirrhinus mrigala* exposed to different concentrations of Furadan at 24, 48, 72 and 96 hrs. exposure (expressed in both percentage and Probit kill after Finney, (1981)

Concentration (mg/l)	Log value of concentration	No. of fish exposed	Fish Mortality									
			24 hr.					48 hr.				
			No. of survival	No. killed	% survival	% kill	Probit kill	No. survival	No. killed	% survival	% kill	Probit kill
CONTROL	-	10	10	0	100	0	-	10	0	100	0	-
10	0.778	10	10	0	100	0	-	10	0	100	0	-
12.5	0.875	10	10	0	100	0	-	10	0	100	0	-
15.6	0.973	10	10	0	100	0	-	10	0	100	0	-
19.5	1.068	10	10	0	100	0	-	10	0	100	0	-
24.4	1.164	10	10	0	100	0	-	-	1	90	10	3.7184
30.5	1.262	10	9	1	90	10	3.7184	6	4	60	40	4.4756
38.1	1.36	10	7	3	70	30	4.4756	2	8	20	80	5.5244
47.7	1.456	10	3	7	30	70	5.5244	0	10	0	100	8.719
59.6	1.553	10	0	10	0	100	8.719	-	-	-	-	-
Concentration factor = 1.25												
			72 hr.					96 hr.				
Control	-	10	10	0	100	0	-	10	0	100	0	-
10	0.778	10	10	0	100	0	-	10	0	100	0	-
12.5	0.875	10	10	0	100	0	-	9	1	90	10	3.7184
15.6	0.973	10	10	0	100	0	-	7	3	70	30	4.4756
19.5	1.068	10	9	1	90	10	3.7184	4	6	40	60	5.2533
24.4	1.164	10	7	3	70	30	4.4756	1	9	-	90	6.2816
30.5	1.262	10	3	7	30	70	5.5244	0	10	0	100	8.719
38.1	1.36	10	0	10	0	100	8.719	-	-	-	-	-
47.7	1.456	10	-	-	-	-	-	-	-	-	-	-
59.6	1.553	10	-	-	-	-	*	-	-	-	-	-
Concentration factor = 1.25												

Average weight of the Fish: 50±3 gm

Temp. 25.3±2.7°

The fish showed characteristic changes in their behaviours such as increasing restlessness, hyper activities, vigorous swimming to the water surface and jumping to out from the toxicant solutions, loss of equilibrium, uncoordinated movements followed by abnormal postures & swimming on one side, coma & death. In few fishes stretched operculum and hemorrhages were also found in the eyes in the fish exposed to higher concentrations of Furadan. The body surface of the fish were comparatively more slimy due to mucus formation and just prior to death, the whole body surface including gill surfaces were covered by mucus secretion, thus obstructing the diffusion capacity of gills; which might be one of the most important factors of death.

Frequent surfacing, jumping & faster opercular movement/frequency were observed in all the fish exposed to higher concentrations of both the pesticides at the beginning, followed by a rapid decline during later stages and prior to death.

Further, the abnormal behaviour of the fish exposed to different concentrations of both Sevin & Furadan, such as increasing restlessness, vigorous swimming on the water surface & jumping outside, hyperactiveness, loss of equilibrium, uncoordinated movements followed by abnormal postures & swimming on one side, coma followed by death, are in agreement with the findings of several workers using various pesticides (Singh *et al.*, 2009, Gupta, 2004; Kumar *et al.*, 2006,) ¹²⁻¹⁴. However, in few fishes exposed to highest / higher concentrations of furadan, stretched operculum and haemorrhages were also recorded.

The toxicity of pesticides based on various forms of Carbamates for fish varies widely as indicated by earlier data (Kumari, 1990) ¹⁵. The 24 hr LC₅₀ value of sevin has been reported to be 1.75 ppm for Longnose Killifish & 4.25 ppm for white mullet (Butler, 1963) ¹⁶ and 6.7 ppm for three-spine stickleback (Steward *et al.* 1967) ¹⁷.

Fish are the most important inhabitants of the aquatic ecosystem both-marine & freshwater in nature. They form one of the most important groups of vertebrates, which influence the human being in various ways. In developing countries like India, millions of people suffer from malnutrition & protein deficiencies. Fish not only provides

protein-rich diet & economical source of food, but also provide a measure source to tide over nutritional deficiencies as their flesh are rich in proteins, vitamins A and D1phosphorus, calcium and several other important elements. In addition, they also provide employment to millions of people all over the world who are engaged in fish culture, fishing, their transport, preservation & marketing. Their export also fetches valuable foreign exchange to the countries engaged in such practices.

The scientific development during the last thirty years or so, have not only provided welfare to the society, but are also responsible for the ecological imbalances, detrimental to living being. Due to rapid industrialization and adoption of new technologies in agriculture works related to many fold increase in the use of pesticides, heavy amounts of industrial effluents & pesticide- residues make their way to surrounding water bodies which ultimately disrupts various ecological balances including hydro-ecosystem. Thus, the fish becomes one of the immediate targets of such pesticides as the fish are more sensitive to biological doses than that of other terrestrial animals.

CONCLUSION

The present investigation entitled, "Mortality of *Cirrhinus mrigala* exposed to different concentrations of Sevin and Furadan" was aimed to find out the impact of toxicity of Sevin and Furadan (Carbamate based pesticides) individually on the fish, *Cirrhinus mrigala* (Ham.) exposed to selected lethal & sublethal concentrations at different periods on the mortality/survival rate of this fish species.

The percent survival of the fish decreased with an increase in concentration of the pesticides and exposure period. The rapid rate of mortality observed, might be due to hyperactivities of the fish and formation of a film of coagulated mucus all over the body surface including gills just prior to death as it would interfere with the respiratory function by obstructing the gas diffusion pathways and/or injury to the gill tissues due to constant exposure of the pesticides.

Effect of Selvin and Furadan on the common Indian Teleostean fishes so as to further the bounds of knowledge in this field, and to make reliable information available to teachers, researchers and students alike.

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Nuvan (DDVP, 2,2-dichlorovinyl dimethyl phosphate) induced biochemical toxicity in an air breathing fish, *Anabas testudineus* (Bloch 1792)

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Abstract- Nuvan, also scientifically known as Dichlorvos is chemically 2,2-dichlorovinyl dimethyl phosphate (DDVP) is an isomer of organophosphate pesticides used commonly to control the insect pest both in standing agricultural field & stored grain products. Nevertheless, the pesticide has adverse effect on the other beneficial organisms such as fishes present in the habitat. This investigation report is related with the toxic effect of Nuvan on the biochemical parameters of an air breathing fish *Anabas testudineus* commonly known as climbing perch having edible value. Different concentrations of Nuvan has been taken for examining the acute toxicity to a fresh water fish *Anabas testudineus* LC⁵⁰ value calculated for the acute toxicity, which was 0.024 ml/L. Biochemical study of kidney Glycogen, Protein profile, Lipid profile, Cholesterol, high density lipoprotein (HDL), low density lipoprotein (LDL) after 24hrs, 48hrs, 72hrs and 96 hrs exposure to lethal toxicity of Nuvan.

Key words: Nuvan, biochemical toxicity, *Anabas testudineus*, organophosphate

INTRODUCTION

Pollution of aquatic ecosystems by extensive and indiscriminate use of toxic chemicals by drift, runoff, drainage and leaching has become one of the most important problems worldwide.¹ Among various toxic pesticides, organophosphates have become the most widely used class of insecticides in the world replacing the low persistent and problematic organochlorine compounds and rapid biodegradability.^{2,3} Dichlorvos (dimethyl-2,2-dichlorovinyl phosphate) is one of the most widely used organophosphorus insecticides used in the control of various pest & ectoparasitic that infect domestic animals,

stored grains and non - target animals of tropical aquaculture. It is also used to combat outdoor and in-home mosquito vectors of several tropical diseases.⁴ The insecticide is highly toxic to fishes and other aquatic organisms. It is a powerful nerve poison, since it inhibits the ACHE (Acetyl choline esterase) activity in the nervous system by blocking synaptic transmission in cholinergic neurons with disruption of the nerve function causing parasympathetic disorders and death of the organism.⁵ Fishes are the most important inhabitants of the aquatic ecosystems which are more frequently exposed to and affected by these toxic pesticides because it is believed that regardless of where the pollution occurs, it will eventually end up in the aquatic environment.^{6,7}

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Pesticides can accumulate in fish and affect human health too via ecological cycling and biological magnification.⁸ The present study is an attempt to estimate the biochemical changes in kidney of fresh water fish *Anabas testudineus* exposed to various concentration of nuvan for different duration of exposure.

Several authors evaluated the toxicity and other effects of dichlorvos as a potential chemical contaminant on various commercially and ecologically important fish species. 96h LC50 values of dichlorvos to different fish species have been reported: *Anabas testudineus* - 2.35mg/L etc.^{9,10,11}

MATERIALS AND METHODS

Collection of the sample species:-

25 fresh samples of the live fish, *Anabas testudineus* were collected from the local fish market of Madhepura, Bihar & kept in the control glass aquarium tank of 20 liter capacity which was sterilized with 1% KMnO₄ & washed two times with non-chlorinated water. The average length and weight of fishes were 10-12 cm and 50-60 g respectively. The control fishes were acclimatized for one week before exposure to the nuvan LC doses. The water used for toxicity test contained 20-25°C and 7.2 pH during acclimations they were fed with market food or egg yolk twice a day. Feeding was stopped 24 hr before starting the experiment. Dead fish (If any) was removed from aquaria as soon as possible to avoid water fouling and water was changed after 2 or 3 days.

Exposure treatments of fishes with different concentration of nuvan:-

Syngenta India Ltd make organophosphate nuvan (DDVP) was used the present study. Four aquaria were set up for each concentration & duration of exposure which contained five fishes in 15 L dechlorinated water. Two different concentrations of nuvan- 2.5ppm & 5ppm were taken in each set of four tanks separately & regular observation on the fishes behavior, morphological changes indicative of biochemical alterations as well as casualty were noticed before taking the blood samples for biochemical assay.

Quantitative biochemical assay:

Blood samples from each treated fish for four different hours of exposure-24 hrs, 48hrs, 72 hrs & 96 hrs were collected 24 hrs after exposure till the death in all

the four tanks. The biochemical parameters were quantitatively estimated by Glycogen, total protein, total lipid, total cholesterol, high density lipoprotein & low density lipoprotein methods respectively.¹²⁻¹⁷ The data was analyzed statistically by log dose/probit regression line method, which was calculated 0.024ml/L. All the fish from each of the control & treated group were anesthetized and the kidney was removed after completion of their exposure time.

RESULTS AND DISCUSSION

From the experiment as conducted above the comparative data on the toxicity of nuvan with respect to different quantitative biochemical assay has been shown below in Table no 1 & 2 that shows a great variation when exposed to different time period. Glycogen content seemed to get reduced with increase in the doses of nuvan during different exposure. Similar finding has also been reported by different authors, they observed loss of glycogen *Channa punctatus*, reported decrease kidney glycogen in *Channa gachua* (Ham.) due to overcome the stress of pollution after M-45 interaction.^{18,19,20}

While protein content also showed great variation in its at different exposure of time. Similar findings was also reported by other authors with decrease protein content in *Channa punctatus* after the exposure of pesticide due to the increased energy demands for the compensation of stress.^{21,22}

As per the picture of biochemical assay showed in the graph I & II (related to Table 1,2) lipid is the only parameter that reflects rising trend in its quantitative concentrations in fishes treated with both the doses of nuvan (2.5 & 5ppm).

The concentration of LDL & HDL reflect most decreasing trend due to nuvan toxicity whereas glycogen & total protein also followed the path of decline. Similar results were also shown with respect to cholesterol, HDL & LDL are synthesized by both liver and intestine and low density lipoprotein is synthesized in the liver only. The present findings also get support with the finding of Lowry and Sargent in fishes who reported the increase lipid metabolism during stress condition. Similar findings were also reported by Sadadamani and Kalamani in *Channa punctatus*. Thus the present study showed that organic pesticides induced alteration in different

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biochemical level such as glycogen, protein, lipid, cholesterol, HDL & LDL depending on different exposure of time.

The alterations in the biochemical parameter also exhibit its corresponding linkage with the overall metabolism & physiology of the fishes which comparatively remain in high order of biological activities in control condition. Therefore, direct or indirect the nuvan

intoxication has adverse impact on the overall biology, metabolism & physiology of fishes in general & *Anabas testudineus* in particular.

The arbitrary use of nuvan in agricultural field is therefore not eco-friendly as it can create adverse impact on the important food resource - fishes present in the crop field or adjoining ponds.

Table 1: Nuvan intoxication (2.5ppm concentration) in biochemical parameters of *Anabas testudineus* during different exposures

Sl no.	Parameters	Control group	Treated group			
			24 hrs	48 hrs	72hrs	96hrs
1	Glycogen	0.9805	0.9731	0.9089	0.8262	0.8092
2	Total protein	0.6570	0.6209	0.5667	0.5087	0.4998
3	Total lipid	1.4311	1.7945	1.7085	1.6954	1.6739
4	Cholesterol	0.5119	0.5066	0.4976	0.4709	0.4535
5	HDL	0.6127	0.5562	0.3291	0.2650	0.1989
6	LDL	0.3460	0.3787	0.1614	0.2240	0.1785

Graph I: Histogram showing the alterations in biochemical parameters of fishes *Anabas testudineus* treated with 2.5ppm nuvan.

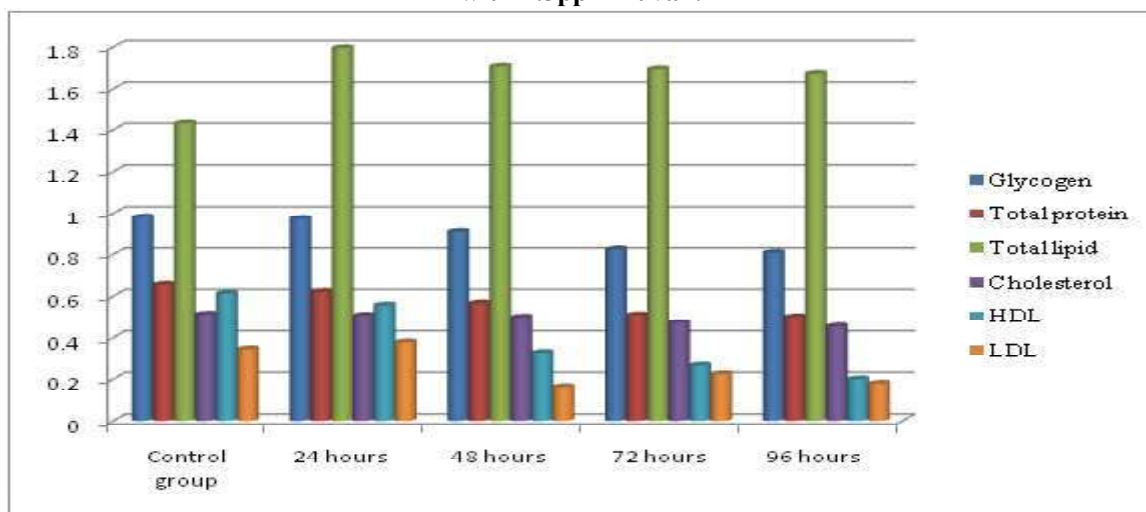
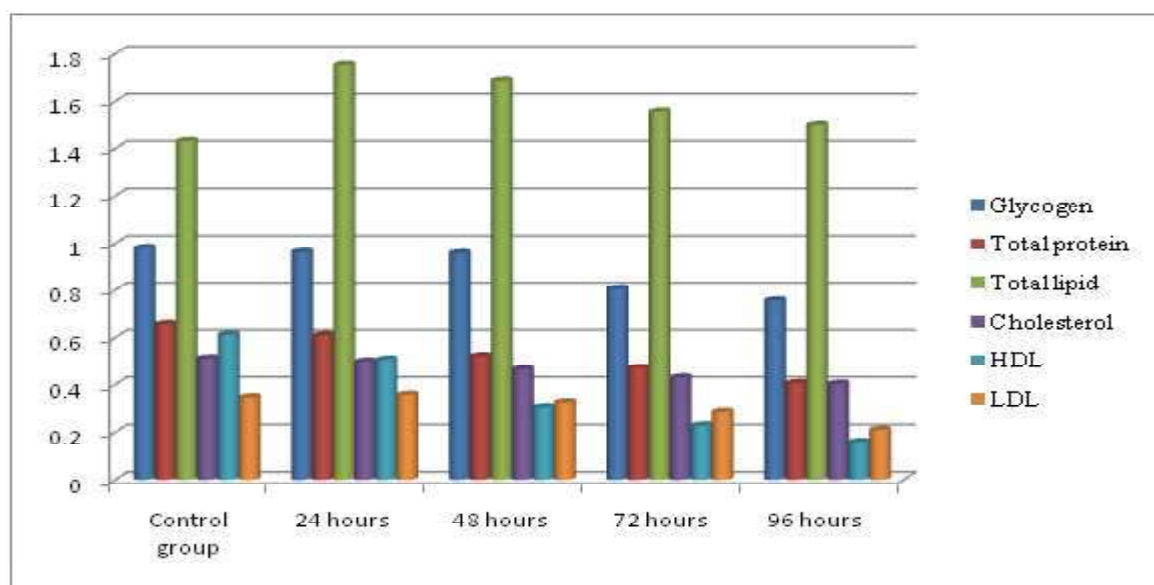


Table 2: Nuvan intoxication (5ppm concentration) in biochemical parameters of *Anabas testudineus* during different exposures

Sl no.	Parameters	Control group	Treated group			
			24 hrs	48 hrs	72hrs	96hrs
1	Glycogen	0.9805	0.9631	0.9589	0.8062	0.7564
2	Total protein	0.6570	0.6101	0.5212	0.4709	0.4065
3	Total lipid	1.4311	1.7542	1.6898	1.6578	1.6023
4	Cholesterol	0.5119	0.4987	0.4655	0.4298	0.4034
5	HDL	0.6127	0.5067	0.3043	0.2286	0.1578
6	LDL	0.3460	0.3568	0.3256	0.2856	0.2098

Graph II: Histogram showing the alterations in biochemical parameters of fishes *Anabas testudineus* treated with 5 ppm nuvan



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Role of carbohydrate in male *Hydrophilous olivaceous* life processes

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Abstract- Carbohydrate metabolism in insects follows a more or less similar pattern as in vertebrates. In insects, like other animals, glucose has a central place in carbohydrate metabolism. But in majority of insect's species, the amount of free glucose is quite small. The most prominent sugar of haemolymph and other body tissues of insect is trehalose, a disaccharide of glucose. It has a glycosidic linkage between anomeric carbon atoms of both glucose molecules. It is a non reducing sugar. In most cases, the sugar contents in the haemolymph is relatively low level of glucose along with higher trehalose contents.¹⁻³ In the present investigation on male *Hydrophilous olivaceous* it was observed that the copper sulphate had an initial carbohydrate lowering effect and latter elevating trend on the haemolymph, fatbody, testis of male insects. Overall initial decline of the carbohydrate content in haemolymph, fatbody, ovary and testes were observed in insects exposed to sub-lethal concentrations of copper sulphate.

Key words: *Hydrophilous olivaceous*, Carbohydrate, sugar, haemolymph

INTRODUCTION

The problem of pollution and its effect on cellular metabolism of an organism is so vital in present complex and civilized society that it has posed a serious threat to purposeful existence of humanity. It has been aptly remarked that during the last decade we have seen a marked increase in the use of toxicants, detergents, industrial effluents, petrochemical products, psychoactive drugs, productive insecticides and other chemicals in our society. It affects all metabolic processes including carbohydrate metabolism directly or indirectly.²

There is no doubt that carbohydrates are the most vital and versatile organic compounds of the living world. These are the most abundant among the major classes of bio-molecules. Chemically they are simple organic compounds that are aldehyde or ketone derivatives of the polyhydric alcohol or as compounds that yield these derivatives on hydrolysis. It is an important source of energy for vital activities. It has central role in cellular metabolism of all living organisms.

Carbohydrate metabolism in insects follows a more or less similar pattern as in vertebrates. In insects, like other animals, glucose has a central place in carbohydrate metabolism. But in majority of insect's species, the amount

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of free glucose is quite small. The most prominent sugar of haemolymph and other body tissues of insect is trehalose, a disaccharide of glucose. It has a glycosidic linkage between anomeric carbon atoms of both glucose molecules. It is a non reducing sugar. In most cases, the sugar contents in the haemolymph is relatively low level of glucose along with higher trehalose contents¹⁻³

Metamorphic changes in both exopterygote and endopterygote are usually accompanied by substantial depletion of their carbohydrate reserves. During this period glycogen and trehalose supply glucose which provides an energy source and the substrate for various metabolic processes. Wyatt (1967) gave a comprehensive account of this process in insects.

Overall initial decline of the carbohydrate content in haemolymph, fatbody, ovary and testes were observed in insects exposed to sub-lethal concentrations of copper sulphate.

MATERIALS

Quantitative estimation of hemolymph, fatbody, ovary or testis carbohydrates of treated and control insects were done by Anthrone reagent Method (Carroll *et. al.*, 1956).

The Anthrone reagent is the basis of a rapid and convenient method for the determination of hexose, aldopentose, hexuronic acid, either free or present in polysaccharides. The blue green solution shows an absorption of maximum at 620 nm. Although some carbohydrates may give another colour.

Reagents

Anthrone reagent: 0.2 % in concentrated H₂SO₄.

Mixture of glucose: 10 mg / 100 ml and glycogen 10 mg / 100 ml (as standard).

Technique

(A) Technique of Haemolymph Carbohydrates

Insect heads were pricked with a needle and were kept inverted in centrifuge tubes, one insect in one centrifuge tube and were centrifuged at 1500 rpm to collect haemolymph. 0.1 ml haemolymph was taken in separate centrifuge tube and added 5 ml TCA to precipitate protein. It was centrifuged at 3000 rpm for supernatant, test for carbohydrates was carried out.

(B) Technique of Fatbody, Testis Carbohydrates

Treated and control insects were dissected. Fatbody and ovaries or testes were taken out and crushed in 2 ml of distilled water with the help of mortar and pestle separately. The crushed material was collected in a centrifuge tube and 4 ml of 10% TCA was added to obtain the precipitate of proteins. It was centrifuged at the 3,000 rpm. The supernatant was kept separately for performing quantitative assay of carbohydrate.

METHODS

4 ml of Anthrone reagent was added to 1 ml of test solution and has rapidly mixed. Boiled in water bath. The absorption was read at 620 nm against the anthrone reagent as blank.

Calculations

The carbohydrate content was calculated as follows -

$$\frac{\text{OD of Unknown Solution}}{\text{OD of Known Solution}} \times \frac{\text{Concentration of Solution}}{\text{Concentration of Known Solution}} \times \frac{\text{Dilution of Known Solution}}{\text{Dilution of Unknown Solution}}$$

Where, OD = Optical Density.

RESULTS AND DISCUSSION

Observation of Carbohydrate Content in *H. olivaceous*

The total carbohydrate content in the haemolymph of mature female and male *H. olivaceous* were 600 ± 10 SD mg / 100 ml and 605 ± 9 SD mg / 100 ml respectively in control insect.

Carbohydrate Content in Mature Male *H. olivaceous* Haemolymph Carbohydrate Content

Almost similar carbohydrate lowering effect on the haemolymph carbohydrate content was observed in mature male *H. olivaceous* on treatment with copper sulphate. At 24 hrs of treatment the carbohydrate content declined to 185 ± 6 SD mg / 100 ml of haemolymph. Similar declining pattern was observed at 48 and 72 hrs after copper sulphate treatment that was 165 ± 4 SD mg / 100 ml and 160 ± 4 SD mg / 100 ml, showing a rising trend, 350 ± 5 SD mg / 100 ml after 168 hrs of treatment. All differences were highly significant as compared it with that of control insects, (Table - 2).

Fatbody Carbohydrate Content

The mature male *H. olivaceous* had a total carbohydrate content of 60 ± 4 SD mg / gram of fatbody in untreated condition or in control insects, (Table - 2).

The carbohydrate content in the fatbody of mature male *H. olivaceous* after different periods of treatment with copper sulphate were 35 ± 3.5 SD mg / gram at 24 hrs, 25 ± 2.5 SD mg / gram at 48 hrs, 16 ± 1.5 SD mg / gram at 72 hrs and 48.5 ± 4 SD mg / gram at 168 hrs. All these differences were highly significant ($P < 0.001$) as compared it with that of control insects, (Table-2).

Testicular Carbohydrate Content

The carbohydrate content in the testis of mature male *H. olivaceous* was 65 ± 5 SD mg / gram in control insects, (Table - 2).

Copper sulphate had also a declining effect on the testis carbohydrate content in the mature male *H. olivaceous*. Carbohydrate content were 17 ± 1.7 ($P < 0.001$) SD mg / gram, 15 ± 1.5 ($P < 0.001$) SD mg / gram, 12 ± 1 ($P < 0.001$) SD mg / gram and 35 ± 2.5 ($P < 0.001$) SD mg/gram of ovary at 24 hrs, 48 hrs, 72 hrs. and 168 hrs respectively after copper sulphate treatment, (Table - 2).

Table - 2 Carbohydrate content in haemolymph, fatbody and testes of mature male *H. olivaceous* during control and after different periods of treatment with Copper sulphate was

Periods of Treatment	Haemolymph mg/100ml	Fatbody mg/gram	Testis mg/gram
Control	605 ± 9	60 ± 4	65 ± 5
24 hrs	$185 \pm 6^{***}$	$35 \pm 3.5^{***}$	17 ± 1.7
48 hrs	$165 \pm 4^{***}$	$25 \pm 2.5^{***}$	15 ± 1.5
72 hrs	$160 \pm 4^{***}$	$16 \pm 1.5^{***}$	12 ± 1
168 hrs	$350 \pm 5^{***}$	$48.5 \pm 4^{***}$	35 ± 2.5

Mean $\pm$ SD

SD = Standard Deviation

* = $P < 0.05$ (Significant)

** = $P < 0.01$ (Significant)

*** = $P < 0.001$ (Significant)

Carbohydrate is structural and functional component of all insect tissues. It is found in nuclei, cytoplasm, and cell membrane as well as in extracellular fluid and supporting tissues. Carbohydrate with proteins and lipids form the principal classes of organic compounds that are found in insect and other organisms. Insects have chitinous exoskeleton which is rich in amino polysaccharides. In addition, insects differ from vertebrates as trehalose, a

disaccharide acts as the storage form of glucose in them. The synthesis and hydrolysis of trehalose is a central hub of carbohydrate metabolism of insects. The main site for the synthesis of trehalose from glucose is the fat body.² In most insects, Carbohydrates are present as glycogen and trehalogen which can readily be converted into glucose. In addition, various amounts of glycoproteins may be present especially in the haemolymph.

Sugar content of whole insect have been studied in some cases; Egorova and Smolin (1962) described a steady rise in trehalose level from egg to pupal stage followed by a decline in the subsequent development to the adult insect in *Antheraea pernyi*. The absorption of glucose through gut wall, its conversion into trehalose in fat cells and finally its distribution in different body tissues were first demonstrated by Kilby (1965) using radio isotopes in locust. In *Bombyx mori*, Nandi (1981) reported increased glycogen concentration in the whole body, fat body and gut during the early period of pupal life which then decreased up to the emergence of the adult from the cocoon. In the gonads, the increasing trend of glycogen continued up to the latter half of the pupal life particularly into ovary.

Marked sex wise variation in the carbohydrate level has been reported for *Tribolium castaneum* and *Coregon cephalonica* (Islam, 1983)¹, *Sphaerodema rusticum* (Sharma and Ehtashamuddin, 1992), *H. olivaceous* (Choudhary, 1996)², *L. maculatus* (Kumari, 1997).

The influence of pesticides and chemicals on the carbohydrate metabolism has been studied in different animals including insects.

In the present investigation on male *Hydrophilous olivaceous* it was observed that the copper sulphate had an initial carbohydrate lowering effect and latter elevating trend on the haemolymph, fat body and testis of and male insects.

Overall initial decline of the carbohydrate content in haemolymph, fat body, ovary and testes were observed in insects exposed to sub-lethal concentrations of copper sulphate.

The carbohydrate content in the haemolymph, fat body and gonads of 25 ± 2 SD mg / gram in testes in control insects. It was 20 ± 2 SD mg / gram in testes at 24 hrs of treatment.

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Population density of zooplanktons of Matashya Beez Prachhetra water body of Dighra, Muzaffarpur (Bihar, India)

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Abstract- As a part of my continued investigation on the planktonic composition of water sample collected from Matashya Beez Prachhetra water body of Dighra, Muzaffarpur which is a fresh water low dimension pond used for fisheries, this article reports on the population density of different zooplanktons found in this aquatic habitat. In addition to several phytoplanktons reported from this water body in my previous research paper, eight different species of zooplanktons belonging to Rotifera, Crustacea & Brachiopoda have been found in this pond with variation in the population density in different season like winter, autumn, summer & rain. The trend of population density & variation in different season has direct correlation with that of phytoplanktons. Surprisingly the density of zooplanktonic species reflects a balance interrelationship with that of phytoplanktons thereby making the pond a steady state system.

Key words: zooplankton population density, Rotifera, Crustacea, Brachiopoda, seasonal fluctuation, Matashya Beez Prachhetra pond, Dighra, Muzaffarpur (Bihar)

INTRODUCTION

All the life forms existing on earth depends upon water for their mere existence & either directly or indirectly affects human beings. Plankton community is considered as one the heterogeneous group consisting of both phytoplankton and zooplanktons population is freshwater aquatic biota that enables to reflect the nature and potential of any aquatic systems.¹

Zooplanktons are sometimes detritivorous plankton drifting in oceans, seas and bodies of fresh water. The word *zooplankton* is derived from the Greek *zoon*, meaning

“animal”, and *planktos*, meaning “wanderer” or “drifter”.² Individual zooplankton are usually microscopic, but some (such as jellyfish) are larger and visible to the naked eye. Zooplankton offer several advantages as indicators of environmental quality in both lakes and rivers. As a group, they have worldwide distribution, species composition and community structure which are sensitive to changes in environmental conditions, nutrient enrichment and different levels of pollution.^{3,4} The pond is regularly used for fishing by the local people besides dumping of domestic solid wastes and waste water. Due to regular dumping of domestic sewage, the pond has become highly eutrophic having dense algal blooms caused by Cyanobacteria. Private entrepreneurs also raise some fish species in this pond.

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The study of zooplanktonic composition abundance and seasonal variation is helpful in planning and successful governing the production of planktonic biomass. A number of workers have reported on different aspect of zooplankton inhabiting Indian fresh waters.^{5,6,7,8,9} It is an important source of aquatic biomass in addition to irrigation. But this water body is under constant threat due to tourist disposal, domestic sewage and increased human activities. It is therefore, urgent need to manage scientifically this water body to tap it maximum potentiality. Here is an attempt that reflects the study of zooplanktonic composition in Matashya Beez Prachhetra Dighra, Muzaffarpur that can provide an effective knowledge about the present condition of the pond.

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MATERIAL & METHODS

The study on zooplanktons were seasonally planned like those of phytoplanktons during the four different seasons winter, autumn, summer & rainy season. Water

samples were collected from different locations of the water body of Matashya Beez Prachhetra from Dighra of Muzaffarpur district during the morning hours 8-11.30 am with the help of plankton net (made of bolting silk of mesh size 20-40 μ m). Each of the samples were collected in plastic bottle and taken in the laboratory for its further study. The labeled concentrated samples were preserved with 1ml of Lugol's solution, simultaneously in 100ml vials.

Identification of sampled phytoplanktons:

The water sample containing the zooplanktons were examined under high resolution stereoscopic microscope (MSZ 20) & the numbers per sample were counted in the plankton counter Sedgewick Rafter slide & identification was done according to the work of various researchers.¹⁰⁻¹⁴ The numbers of each group of phytoplanktons were recorded in the observation Table no. 1 also showing the seasonal variation.

Statistical Analysis

Analysis of relative abundance of zooplanktons with respect to various numbers of sampling being the observation number of collected data had also been done by using the relative abundance statistical formula, $p_i = \frac{n}{N} \times 100$, where p_i is i^{th} number of zooplanktonic species found, n = number of individuals of a species, N = total number of individual of all the species. The statistical result has also been reflected in bar diagram or histogram as shown in graph 1. For the statistical analysis of these phytoplanktons the standard methods given by different scientists were used.¹⁵



Fig 1 :-Phytoplankton net(mesh size:20-40 μ m)

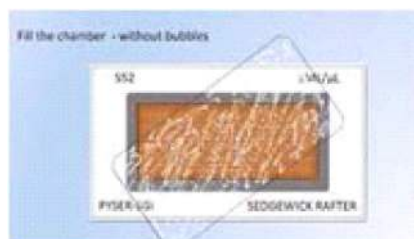
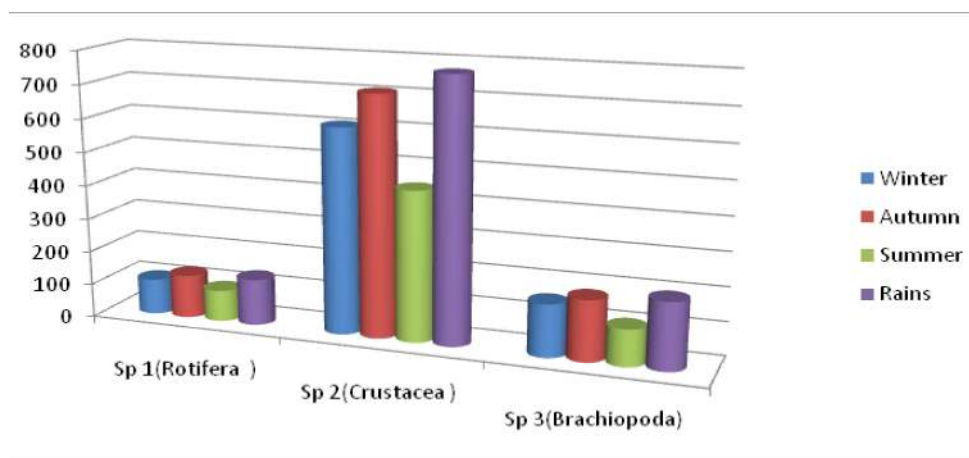


Fig 2: Sedgewick Rafter plankton counter

Table 1: Seasonal mean population & relative abundance of zooplanktons sampled from Matashya Beez Prachhetra water body of Dighra

Seasons	Winter		Autumn		Summer		Rains	
Group & genus	Individual sample population per SNS	Relative abundance % (N/n*100)	Individual sample population per SNS	Relative abundance % (N/n*100)	Individual sample population per SNS	Relative abundance % (N/n*100)	Individual sample population per SNS	Relative abundance % (N/n*100)
(A) Rotifera								
(i) <i>Keratella sp.</i>	81	77.14	98	76.56	75	80.64	105	74.80
(ii) <i>Brachionus sp.</i>	24	22.85	30	23.43	18	19.35	32	325.19
Total	105		128		93		137	
(B) Crustacea/Copepoda								
(i) <i>Cyclops</i>	100	16.42	120	16.92	70	15.73	130	17.56
(ii) <i>Mesocyclops</i>	68	11.16	79	11.14	45	10.11	88	10.13
(iii) <i>Eucyclops agilis</i>	123	20.19	130	18.33	100	22.47	156	18.24
(iv) <i>Diaptomus sp.</i>	318	52.21	380	53.59	230	51.68	400	54.05
Total	609		709		445		774	
(C) Brachiopoda/Chladocera								
(i) <i>Daphnia sp.</i>	92	58.97	105	58.33	64	59.25	120	60.60
(ii) <i>Ceriodaphnia sp.</i>	64	41.02	75	41.66	44	40.74	78	39.39
Total	156		180		108		198	
Grand total	870		1017		646		1088	

Graph 1: Zooplanktonic composition of Matashya Beez Prachhetra water body of Dighra



RESULTS & DISCUSSION

The population density of zooplankton species found in the Matashya Beez Prachhetra water body taxonomically belong to three groups of lower invertebrates like Rotifera, Crustacea and Brachiopoda. Rotiferans are represented by the two taxa- *Keratella sp.*, *Brachionus sp.*, Crustaceans by four - *Cyclops*, *Mesocyclops*, *Eucyclops agilis*, *Diaptomus sp.* and Brachiopoda by two species - *Daphnia sp.* & *Ceriodaphnia sp.* with the significant variation in their relative abundance indicating population density. Group wise comparison as shown in above table reflects

that crustaceans were among the most populated species with respect to rotifers & brachiopoda. As a matter of fact, relative abundance of species is a statistical dimension of numerical value of mean population density of any species. plankton since the 1930s with numerous research projects continuing today

The finding presented in the tables & graphs in this article reflect the individual mean population per quadrate of standard aquatic net sweep observed during collection of the random sample of agitated water from different parts of the above pond. The data also indicate significant variation in the population density of zooplanktons in

different seasons of the experimental year 2016-2018. It is clear from the graphics prepared from the tabular data of sampled zooplanktons that the overall seasonal mean population density of zooplanktons was found to be higher during the rainy season followed by autumn & winter season but minimum during the summer season. This gradient of seasonal fluctuations in zooplanktonic population density is directly related with the varying water temperature during the seasons & also the changing profile of pond's physicochemical property. The other reason of seasonal fluctuation may be correlated with the variation in the phytoplanktonic as well as other faunal population existing at different trophic levels in the pond.

The synergistic consideration of fluctuating zooplankton population density along with that of phytoplankton derives important information regarding the biological health of this pond that it is eutrophic in nature observing recycling of different mineral, abiotic & biotic resources to sustain the diversity & density of various life forms like zooplanktons, phytoplanktons, fishes and insect larvae.

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Demographic pattern of prevalence of intestinal protozoan infections with special reference to schedule caste and minorities population of Patna District of Bihar, India

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Abstract- Intestinal parasitic infections are endemic to worldwide and remain a major public health problem among most communities in tropical and subtropical countries including India. This study was carried out in the population of some ghettos (slums) of Patna District of Bihar, India during 2010-2011 with the purpose of achieving a better understanding of the incidence and factors associated with intestinal protozoan parasites among the population of selected ghettos. Stool samples from 687 subjects were collected randomly from the population and examined by simple saline preparation technique. In addition, stained preparation by Lugol's iodine solution was also used for the identification of nuclear character of the cysts. (19.6%) of the total examined cases (135/687) were found to be infected with various protozoan parasites. The species specific degree of incidence for the protozoan parasites has been found to be- *Entamoeba histolytica* (8.0 %), *Giardia lamblia* (5.0%) and *Entamoeba coli* (4.3%). About 2.1% of the infected samples contained multiple infections. Samples were collected from two different communities, schedule caste (SC) and Muslims. 70 samples out of 310 were positive for parasitic infections in SC groups which constitute 22.5% of infection rate, in Muslims 65 (17.2%) were harbored with parasitic infections out of 377 samples studied. The present study revealed that the prevalence of intestinal protozoan parasitic infections among the population of the areas of Patna is still so high that required special control measures. Furthermore the study underscored the need for surveillance and intervention program to control and manage these infections.

Key words: Intestinal protozoan parasites, Schedule Caste (SC), Muslims, Prevalence, Ghettos, Patna district

INTRODUCTION

Parasitic infections caused by intestinal protozoan parasites are globally endemic and it represents one of the major health problems in the developing countries with high prevalence rate in many regions including India.¹ These infections remain significant cause of morbidity and mortality along with other manifestations like iron deficiency anaemia, growth retardation in children, and

other physical health problems.² 60% of the world population is infected with abdominal parasites.³ WHO estimated approximately 50 million people around the world suffer parasitic infections each year, resulting mortality rate 40-100 thousand per year.⁴⁻⁶ About 39 Disability Adjusted Life Year (DALY) associated with parasitic infection presents a major health burden.⁷ The most common intestinal protozoan parasites are *Entamoeba histolytica*, *Entamoeba coli* and *Giardia lamblia*. The disease caused by these parasites is known as giardiasis, amoebiasis and they are associated with diarrhoea.⁸ *Giardia lamblia* is the most prevalent parasite

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which causes diarrhoea in developed world as well as in developing countries. Amoebiasis is the third leading cause of death from parasitic disease in worldwide. The prevalence of parasitic diseases depends on environmental, social and economic factors.⁹ Besides this poverty, illiteracy, unhygienic surroundings, lack of safe potable water, moreover some environmental factors such as global warming, subtropical, hot and humid climatic conditions, pollution, urbanization and unplanned reforestation may be the reasons for the transmission of parasitic infections.^{10,11}

In India a wide range (11.40 – 90.70%) of community based prevalence of intestinal parasitic infections has been reported from different geographical provinces and the populations.¹²⁻¹⁴ The prevalence of intestinal parasitic infections varies in different geographical regions.¹⁵ Thus it is important to know the disease burden of the parasitic infections in the community level. Scant information is available in this regard especially in Bihar, whereas a study related to overall prevalence of intestinal parasitic infections have been published in recent past year.¹⁶ So we performed a community based study especially in Schedule Caste (SC) and Muslims. These clusters of populations are regionally or locally known as ghettos (Slum dwellers, a part of a city, especially slum area occupied by a minority group or groups). The main purpose of the study was to evaluate the incidence of the intestinal protozoan parasites in the population under study.

MATERIALS AND METHODS

A cross sectional study (2010- 2011) was carried out in the some ghettos of Patna district of Bihar, India for the prevalence of intestinal protozoan parasitic infections. SC and Muslim communities reside in these areas which are socio- economically very poor and backward. A total no. of 687 stool samples was randomly collected, out of which 310 samples were from SC and 377 samples from Muslims. Data was collected using a pre structured questionnaire to obtain the socio- demographic profile and a sterile, dry leak proof, clean glass container with and without 10% formalin bearing necessary code. Other details regarding age, sex, religion, and food habits, living status, source of drinking water and laboratory facilities were also recorded. First of all a macroscopic examination of all the collected samples was carried out for color, consistency and presence

of mucus and blood. For microscopic examination, Saline wet mount and Lugol's iodine wet mount were prepared as per standard protocol for the identification of eggs, cysts and trophozoites of intestinal parasites.¹⁷

RESULTS

The result of the overall survey has been summarized in Fig 1. Out of a total 687 fecal samples examined 135 (19.6%) of individual were found to be infected with intestinal protozoan parasites. In sex related incidence of infections, males showed slightly higher rate (20%; 84/420) than females, (19.1%; 51/267). Out of 310 samples of SC, 70 (22.6%) samples were found infected. Regarding sex wise, males (25.7%; 50/194) showed higher infection rate than females (17.2%; 20/116). Out of 377 samples of Muslims 65 (17.2%) samples were found infected. Gender wise females (20.5%; 31/151) showed higher infection rate than males (15%; 34/226) (Table 1). Species wise *Entamoeba histolytica*, (8.0%) was most prevalent followed by *Giardia lamblia* (5.0%) and *Entamoeba coli* (4.3%) multiple infections (2.1) (Table 2).

DISCUSSIONS

Intestinal parasitic infections are devastating and prevalent in tropical and subtropical countries and constitute a major health problem in many developing countries predominantly due to poor sanitation and inadequate personal hygiene. Intestinal parasitic infections including protozoan infections remain a major threat to health, because chronic conditions may lead to serious disease and have been described as constituting the greatest single worldwide cause of illness and disease.¹⁸ The present study showed that out of 687 individuals 135 (19.6%) were harbored with various species of parasites and three species (Table 3) of parasites were detected. In this study 310 samples were collected from SC category and 377 from Muslims out of which 70 (22.5%) and 65 (17.2%) samples respectively found to be infected. These results were consistent with a wide range of (11.5% - 97.4%) prevalence of parasitic infection has been reported from different geographical regions and populations of India.¹⁹⁻²¹ The high rate of infection may be correlated with various factors such as behavioral, biological, environmental socio economic and health system. Beside this poor sanitation, lack of safe drinking water, open air defecation, barefooted

Begum- Demographic pattern of prevalence of intestinal protozoan infections with special reference to schedule caste and minorities population of Patna District of Bihar, India

walking in the field pattern of setting of life style, warm and humid climate and lack of health education in the study area. In sex related incidence of infection in overall prevalence and in SC groups males (20% and 25.7%) showed slightly higher infection rate than females (19.1% and 17.2%). These results are similar to those reported in previous study¹⁴, but they are not in line with those obtained by other authors who found that the females show high infection rate.²²⁻²⁴ High rate of infection in males may be attributed to their maximum daily life activities, greater exposures to worldly environment and also for their careless living habits. In Muslims communities' females (20.5%) show higher infections rate than males (15%). Higher prevalence among females in Muslims may be due to their prolonged contact with prevailing unhygienic surroundings, low literacy rate, lack of knowledge of proper hygiene and carelessness about health, most of the Muslims localities are situated in sub hygienic areas and lack of awareness also contribute for the prevalence of infections. With respect to the different species of protozoan *Entamoeba histolytica* showed a relatively higher incidence than *Giardia lamblia* and *Entamoeba coli*. This result is similar to those reported in other parts of

India including Bihar^{6,19,22} but they are not in consistent with those reported by other authors who found that *Giardia lamblia* was the most prevalent protozoan species.²⁵ The high rate of protozoan in our study could be due to the existence of resistant cysts of the parasites in the study area. Therefore the present study confirms that occurrence of protozoan infections are quite high in the study population. The remoteness of the rural settings, characterized by the absence of required infrastructure like safe drinking water, toilets, medical facilities and basic sanitation and inadequate environment constitute the main determining factors that predisposed this population to parasitic infections. This therefore highlights the need to educate members of the community and encourage hygiene practice especially those in the study areas, on the method of preventing transmission and spread of these intestinal protozoan parasites. However mild the infection may be, it does hamper the nutritional status, working ability and consequent economic and educational development of the individual concerned. This becomes much more imperative when the majority of population of the country is younger ones.

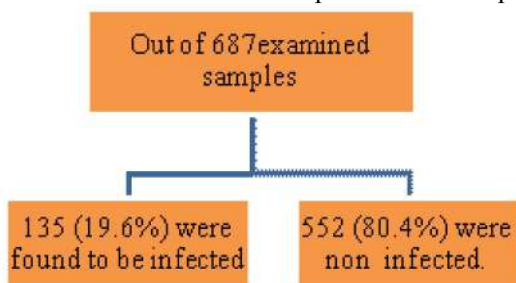
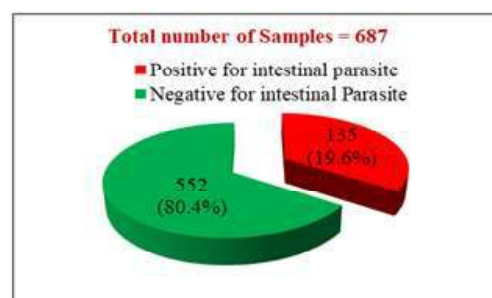


Figure 1.



Pie Chart with reference to Figure 1

Table 1. The incidence of intestinal protozoan parasitic infections of the population studied.

Variables	Total No. of samples Studied	No. of infected samples	Prevalence (%)
(i) Overall	687	135	19.6
Male	420	84	20.0
Female	267	51	19.1
(ii) Schedule Caste	310	70	22.5
Male	194	50	25.7
Female	116	20	17.2
(iii) Muslims	377	65	17.2
Male	226	34	15.0
Female	151	31	20.5

Table 2. Species wise prevalence of protozoan infections out of 687 samples studied.

Parasites	No. of infected samples (135)	Prevalence (%)
<i>E. histolytica</i>	55	8.0
<i>G. lamblia</i>	35	5.0
<i>E. coli</i>	30	4.3
Multiple infections	15	2.1

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Portfolio function of lady bird beetles (Coccinellids) in the agro forestry ecosystem of Kanke block, Ranchi district (Jharkhand), India

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Abstract- Portfolio function of any biotic population in an ecosystem has multidimensional control & it is primarily engineered by S-W diversity index of the species. S-W diversity index is popular statistical information index as quantitative tool to assess the species abundance, richness & bifold product value of group of species dwelling together in a habitat that yields significant result for understanding their asset & liability value in the portfolio of biotic community. It further helps in contemplating the strategies for the management & development of sustainable environment & ecosystem. Beetles sampled from the forest, agricultural areas & grasslands of Kanke block in Ranchi of Jharkhand were taxonomically identified as *Coccinella septempunctata* L., *Brumus suturalis*, *Adalia decempunctata* Linn., *Cycloneda sanguinea*, *Cryptocephalus triangularis*, Hope & *Hippodamia parenthesis*. When the data were subjected to S-W diversity index tool, the value of H ($H = -\sum p_i \log p_i$) for all the species of beetles was found to be 0.7691, 0.7625 & 0.7616 and PF (Portfolio Index) as 104.0, 19.67 & 78.7 which indicates that they have profitable asset investment in the community. as per portfolio function laws.

Key words: S-W diversity, lady bird beetles, profitable asset investment, portfolio function laws, agroforestry ecosystem

INTRODUCTION

Insects are one of the most diversified group of fauna found in over all parts of the world.¹ Insects pests constitute a very serious problem in our country. According to Sir Walter Morley Fletcher “at a very modest estimate, the losses caused to India by insects that attack crops, timber and animal products cannot be estimated as less than 200 crores of rupees and a loss of over a million and a half of human lives”. Four types of defense mechanism such as mechanical, cultural, chemical and biological has been developed by the entomologists to prevent the injurious insects becoming abundant. Hence the biological control

is based on the assumption that number of insect pests is usually destroyed by insect only acting as predators or parasites. Family Coccinellidae are among those group which acts as predators for most of the insects that come under the order Coleoptera. These are a group of small beetles that usually ranges in size 0.8 to 18 m.² In Northern America this family is known as ladybugs & in Britain and other parts of the world it is known as ladybirds. Usually the entomologists call this insect as ladybugs or ladybeetles since these insects are not classified as true bugs.³

Thus the family in which these insects belong are very diverse in habitat & are found in all types of terrestrial

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ecosystem. They are one of the most important group of the natural predatory of many horticultural and agricultural crop pests such as aphids or scale insects.^{4,5} Many coccinellids have been seen laying their eggs directly in aphid and scale insect colonies in order to ensure their larvae have an immediate food source.⁶ Coccinellidae are generally yellow, orange, or red in colour with small black spots on their wing covers, with black legs, heads and antennae. However, much variation has been seen in their colour pattern and spot. For example, a minority of species, such as *Coccinella septempunctata* L have seven black spots on their back while *Vibidia duodecimguttata*, is twelve-spotted species, have whitish spots on a brown background. Coccinellids are found worldwide, with over 6,000 species described and dived into six subfamilies: Sticholotidinae, Chilocorinae, Scymninae, Coccidulinae, Coccinellinae and Epilachninae.^{7,8} Here in this paper attempt has been made to study on the portfolio function of the lady bird beetle in forest, agricultural & grassland areas of Kanke since these

insects are eco friendly in nature their availability is very important for maintaining balanced ecosystem.

MATERIAL AND METHODS

For the study of portfolio function of Coccinellidae, collection of the samples were done from agro forestry and grassland habitats of Kanke block present in Ranchi district which is situated in north-eastern part from the main city consisting of 90 acres of land under forest ecosystem that is integrated into agricultural management systems. Specimens were collected by using net, hand picking and light trap; one light trap per 10 unit quadrates for all the three sites. These collected species were then identified using Imm's book of entomology & Modern Entomology. Study was conducted from three different sites & was visited once in every fifteen days. The data collected from the above findings have been given in table number 1, 2 & 3. On the basis of the procured data, species diversity by S-W index & portfolio function index (PFI) by linear dividend method were calculated.

Table 1: Portfolio function of lady bird beetle in crop habitat (paddy, wheat, gram, moong)

Sl. No.	Names of lady beetles sampled	Avg. number of individuals per unit quadrate (n)	individual abundance pi(n/N)	Relative abundance % (pi=n/N*100)	log pi	pi*log pi	H	Per quadrate average no. of host plants (paddy, wheat, gram, moong)	PF index = per quadrate avg. no. of host plants/diversity value of lady beetles
1	<i>Coccinella septempunctata</i> L	60	0.218	21.8	-0.661	-0.1440	0.7656	80	80/0.7656=104.4
2	<i>Brumus suturalis</i>	49	0.178	17.8	-0.749	-0.1333		80	
3	<i>Adalia decempunctata</i> Linn.	30	0.109	10.9	-0.962	-0.1048		80	
4	<i>Cycloneda sanguinea</i>	48	0.174	17.4	-0.759	-0.1320		80	
5	<i>Cryptocephalus triangularis</i> Hope	52	0.189	18.9	-0.718	-0.1357		80	
6	<i>Hippodamia parenthesis</i> Say	36	0.130	13.0	-0.886	-0.1158		80	
	Grand Total (N)	275							

Table 2: Portfolio function of lady bird beetle in forest area (Acacia, Peepal, Gamhar)

Sl. No.	Name	Avg. number of individuals per unit quadrate (n)	individual abundance pi (n/N)	Relative abundance % (pi=n/N*100)	log pi	pi*log pi	H	Per quadrate average no. of host plants (Acacia, Peepal, Gamhar)	PF index = per quadrate avg. no. of host plants/diversity value of lady beetles
1	<i>Coccinella septem punctata</i> L	45	0.211	21.1	-0.675	-0.1424	0.7625	15	15/0.7625=19.67
2	<i>Brumus suturalis</i>	31	0.145	14.5	-0.838	-0.1215		15	
3	<i>Adalia decempunctata</i> Linn.	20	0.093	9.3	-1.031	-0.095		15	
4	<i>Cycloneda sanguinea</i>	35	0.164	16.4	-0.785	-0.1287		15	
5	<i>Cryptocephalus triangularis</i> Hope	42	0.197	19.7	-0.705	-0.1388		15	
6	<i>Hippodamia parenthesis</i> Say	40	0.187	18.7	-0.728	-0.1361		15	
	Grand Total (N)	213							

Kumari-Portfolio function of lady bird beetles (Coccinellids) in the agro forestry ecosystem of Kanke block, Ranchi district (Jharkhand), India

Table 3: Portfolio function of lady bird beetle in grassland (small wild herbs, grasses, puttus, bhangaria)

Sl. No.	Name	Avg. number of individuals per unit quadrat (n)	individual abundance pi (n/N)	Relative abundance % (pi=n/N*100)	log pi	pi*log pi	H	Per quadrat average no. of host plants (small wild herbs, grasses, puttus, bhangaria)	PF index =per quadrat avg. no. of host plants/diversity value of lady beetles
1	<i>Coccinella septempunctata</i> L.	40	0.220	22	-0.657	-0.1445	0.7616	60	60/0.7616= 78.7
2	<i>Brumus suturalis</i>	23	0.127	12.7	-0.896	-0.1137		60	
3	<i>Adalia decempunctata</i> Linn.	19	0.104	10.4	-0.982	-0.1021		60	
4	<i>Cycloneda sanguinea</i>	28	0.154	15.4	-0.812	-0.1250		60	
5	<i>Cryptocephalus triangularis</i> Hope	33	0.182	18.2	-0.739	-0.1344		60	
6	<i>Hippodamia parenthesis</i> Say	38	0.209	20.9	-0.679	-0.1419		60	
	Grand Total (N)	181							

The host plants on which beetles were found wriggling freely have also been identified & recorded to consider the available food resource for the beetles in the selected habitats (cropland, agricultural areas & grasslands)

RESULT AND DISCUSSION

The concept of 'portfolio function' is completely based on financial management of different types of fund investment fetching various returns in the terms of yields. The term function in portfolio management is the ultimate result of asset and investment added to liability expenses in a specific slot of time & space. When this concept is applied to ecological research, it provides an important insights into how ecosystems are organized in different habitats & how biotic species are interrelated in terms of autotrophs-heterotrophs, producers-consumers & managers-employees relationships. The holistic trends of evolutionary strategies of any community emerge on these parameters which predict the development of the environment & nature. It can also help in providing appropriate measures for developing management and conservation schemes, and offer an approach that does not rely on prescriptive predictions about threats in an uncertain time scale.

In the present investigation, 6 different species of lady bird beetles from three different habitats were sampled during the experimental year 2018-2019 & identified as- *Coccinella septempunctata* L., *Brumus suturalis*, *Adalia decempunctata* Linn., *Cycloneda sanguinea*, *Cryptocephalus triangularis* Hope & *Hippodamia parenthesis*, which were present in different degree of abundance depending upon the availability of various host plants as food resources. In habitat 1 (crop land), where large number of paddy, wheat, gram & moong plants were

available, the S-W diversity value of all the six lady beetle species have been found to be 0.7656 sponsored by average 80 number of plants per 10 unit quadrates as food resource. Hence the portfolio function index of these beetles in relation to the host species was 104.4. Similarly in habitat 2 (agricultural area), where average number of Acacia, Peepal & Gamhar trees were available, the S-W diversity value of all the six lady beetles species have been found to be 0.7625 sponsored by least 15 number of trees per 10 unit quadrat as food resource. Hence, the portfolio function index of these beetles in relation to the host species was 19.67. As well as in habitat 3 (grassland area), where number of small wild herbs, grasses, Puttus, Bhangaria were available, the S-W diversity value of all the six lady beetles species have been found to be 0.7616 sponsored by average 60 number of plants per 10 unit quadrat as food resource. Hence, the portfolio function index of these beetles in relation to the host species was 78.7.

Comparatively the performance of these six lady beetle species as reflected by different values of habitat specific portfolio function index which was highest in habitat 1 followed by habitat 3 while it was lowest in habitat 2. Therefore, the habitat 2 remains to be more properly managed in terms of host plants & the predatory beetles whereas the remaining two habitats enjoy comfortable & rich environmental ecosystem leading to good production of crops & useful grassland species where the predatory beetles remained busy in destroying the harmful pests of the plants & scavenging the undesirable pathogens of the habitats. Thus computing the portfolio function index on the basis of S-W diversity index of any group of species in relation to the habitat resources provides significant indication about the wellbeing of the habitat.

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